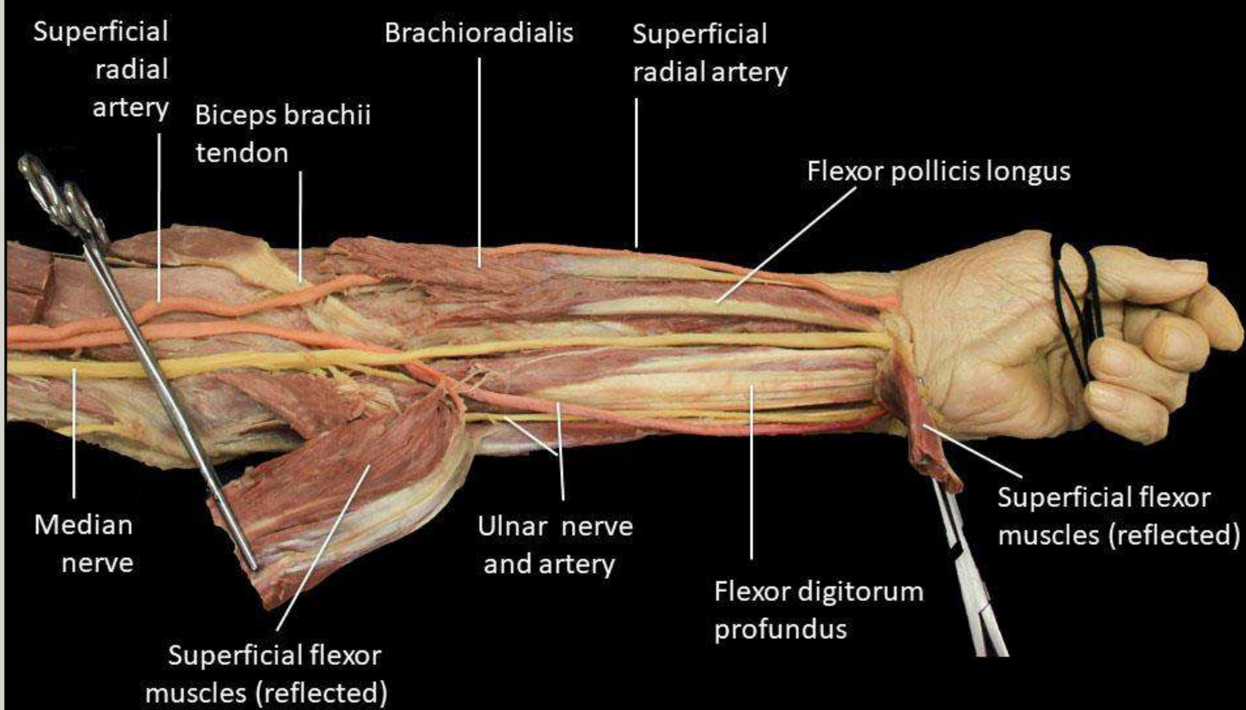
ANATOMY

The Upper Limb

The Forearm



Dr. Rasha Alshali

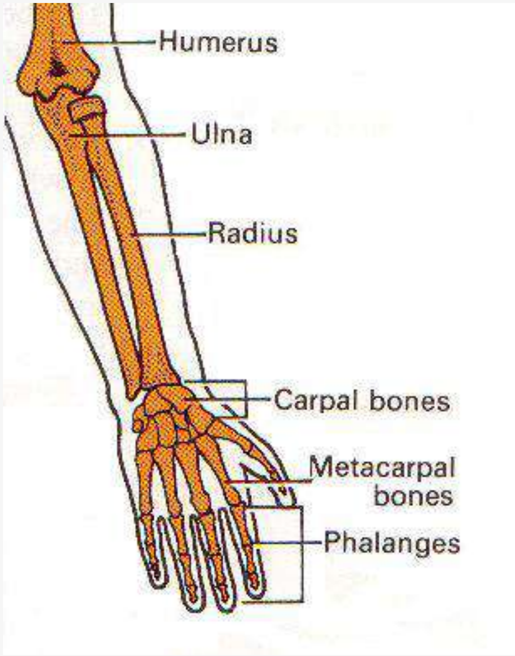
Bones of the forearm

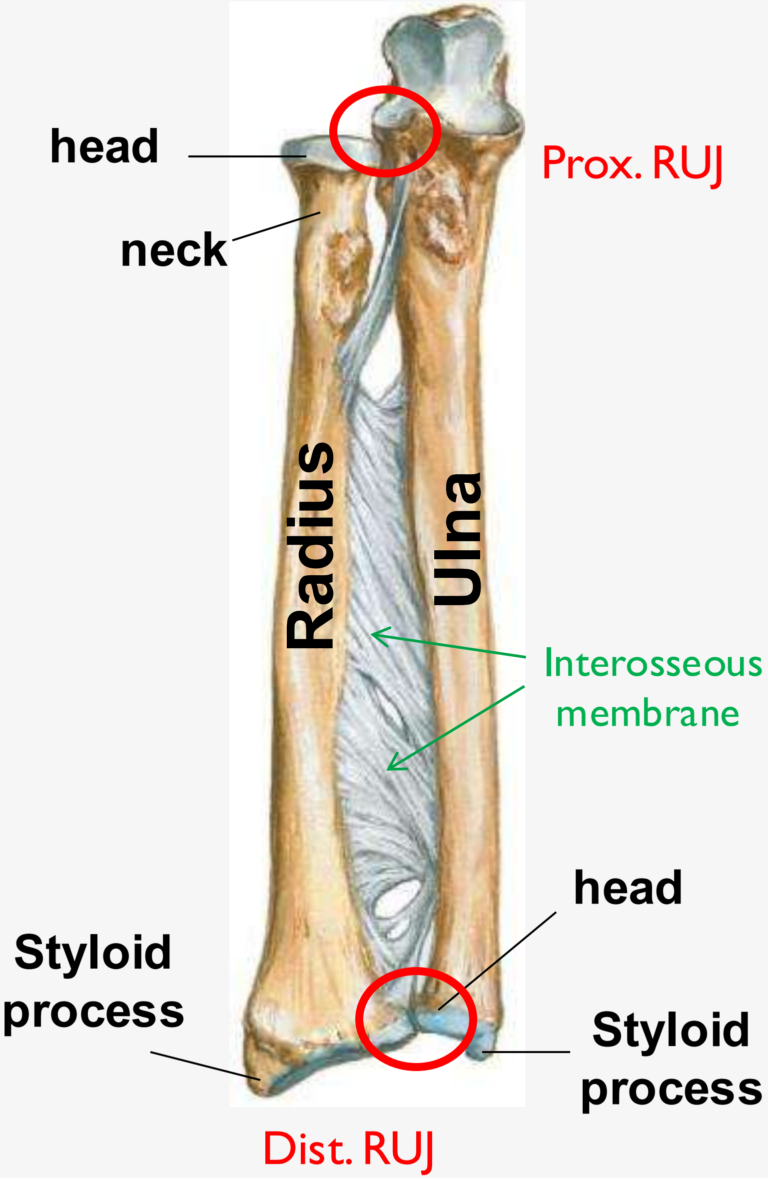
Forearm:

- **Radius**, laterally
- **Ulna**, medially

Hand:

- **Carpals** (8 bones)
- **Metacarpals** (5 bones)
- **Phalanges** (3 in each finger, except the thumb only 2 phalanges)





Bones of the Forearm

Radius

The **lateral** bone of the forearm

Features:

Upper/ proximal end (**head**)

Below the head → **neck.**

Below the neck → **radial tuberosity.**

The shaft has borders & surfaces

interosseous border medially for the attachment of the interosseous membrane.

Lower/ distal end has Ulnar notch medially

And **styloid process,** projecting laterally

Articulations:

-Its proximal end (head) articulates with:

‘humerus (capitulum)

‘ulna (radial notch of ulna)

-Its distal end articulates with:

‘scaphoid & lunate bones (inferiorly)

‘ulna (medially)

Bones of the Forearm

Ulna

The **medial** bone of the forearm

Features:

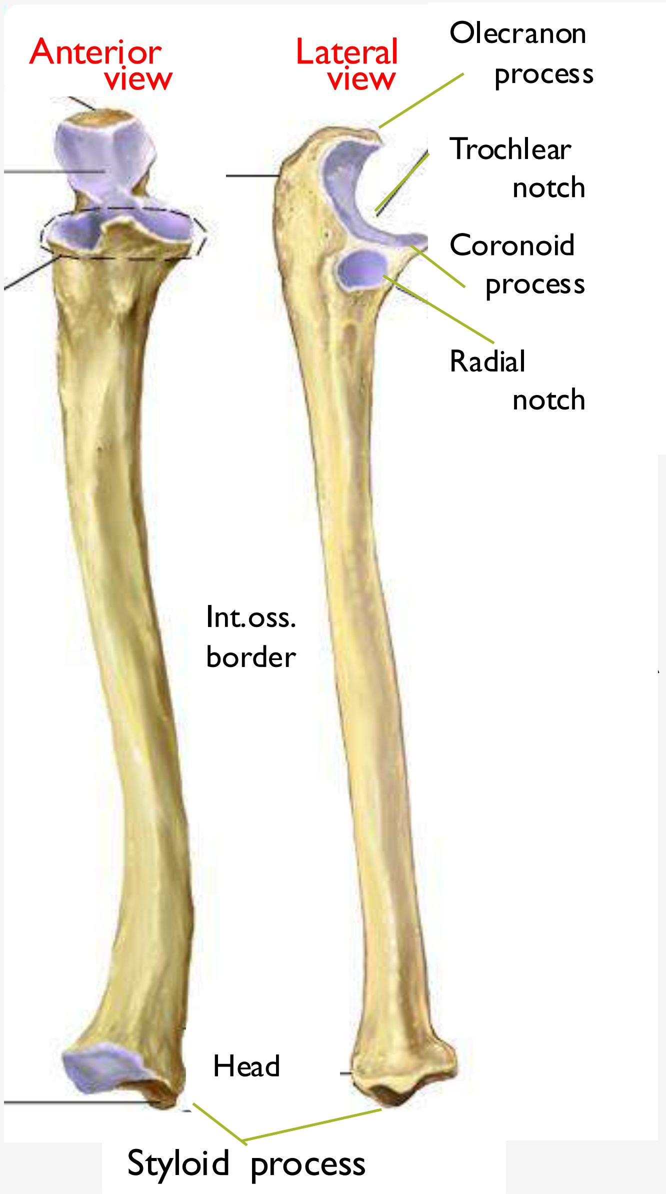
Upper/ proximal end is large, it shows:

- The **olecranon process** (forms 'prominence of 'elbow).
- Anteriorly, the **trochlear notch,**
- The **coronoid process,** on its lateral surface
  the **radial notch** articulates with 'head of 'radius.

The **shaft** has borders & surfaces:

- The **interosseous border** is sharp & **laterally**
- The posterior border is rounded & subcutaneous

Lower/ distal end shows a small rounded **head,**
and the **styloid process,** projecting **medially**.



Articulations:

Proximally, ulna articulates with:

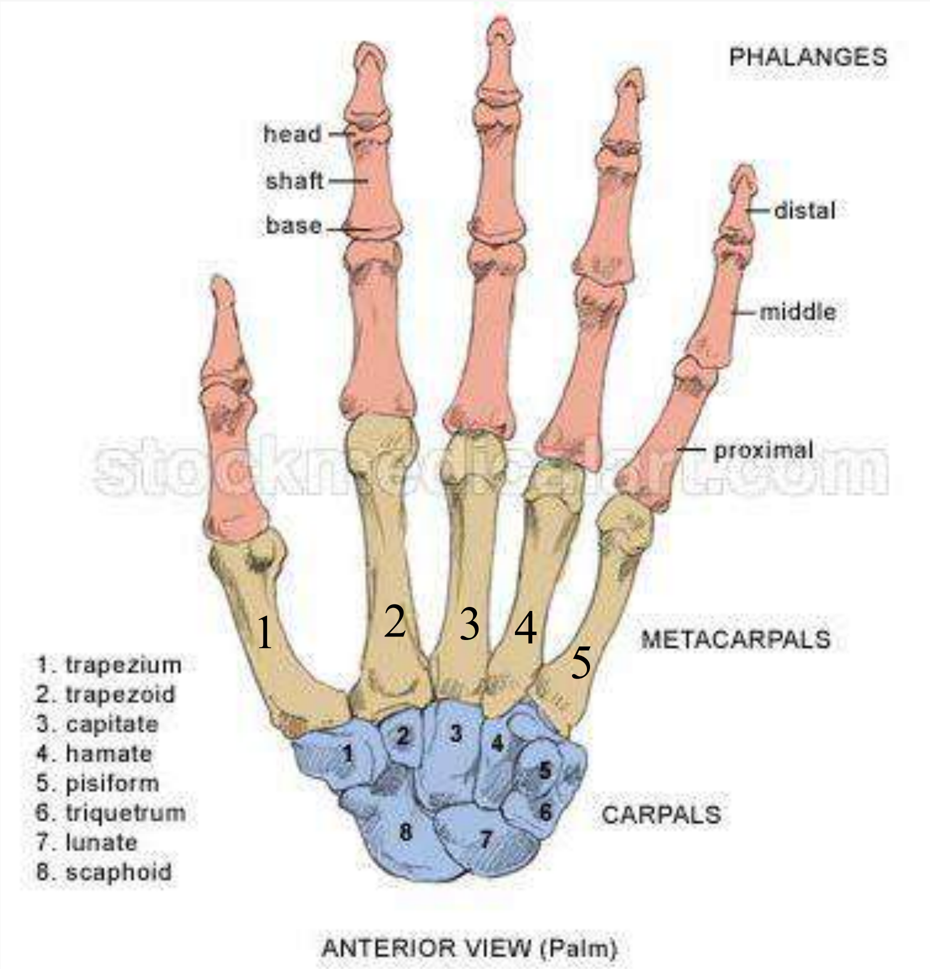
- The humerus
- The radius

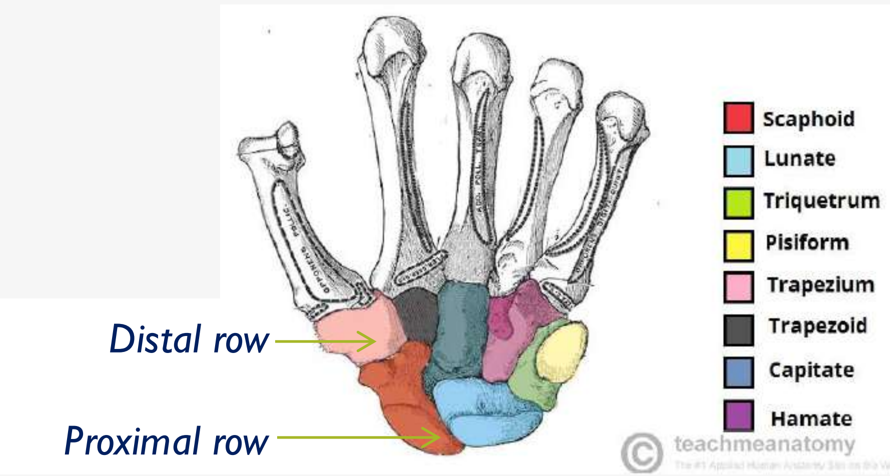
Distally, ulna articulates with:

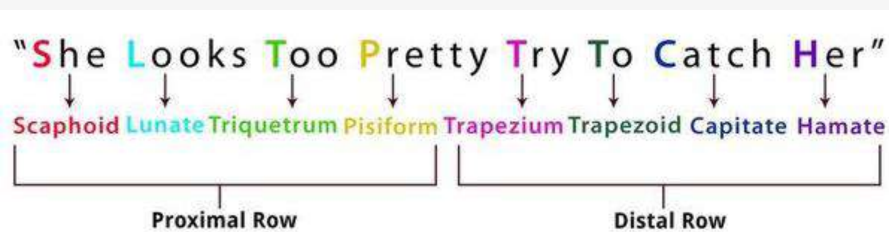
'radius (laterally)

NO articulation with carpal bones

Bones of the Hand







From lateral to medial

CONTENTS OF LECTURE:

Bones of the forearm → Practical

Muscles of the forearm (Flexors & Extensors)

- ✓ Anterior , posterior & lateral compartments

Flexor & Extensor retinaculum

- ✓ Attachments
- ✓ Structures passing deep & superficial to 'retinaculum

The arteries: Radial and ulnar arteries

- ✓ The course, branches, main relations
- ✓ Site of pulsation (arteries).

The nerves: median , radial and ulnar nerves

- ✓ The course, branches, main relations

Anatomic snuffbox:

Boundaries & contents

Carpal Tunnel Syndrome:

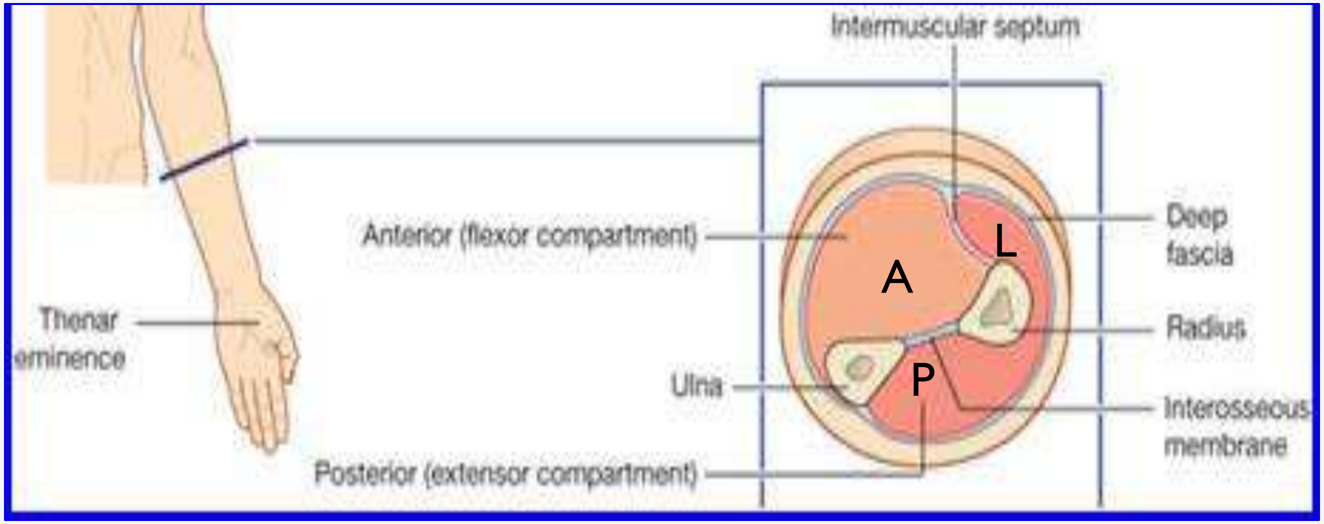
Cause & treatment

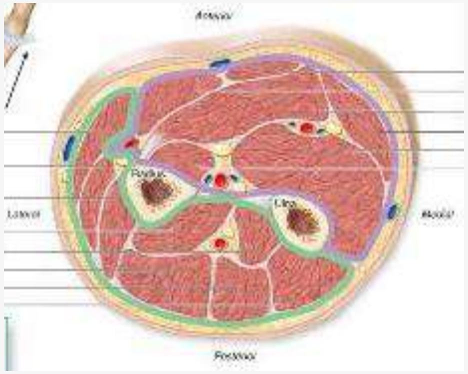
Fascial compartments of the forearm

The forearm is divided into facial compartments by:
the surrounded sheath of deep fascia,
the interosseous membrane and fibrous intermuscular septa

So, the forearm is divided into:
Anterior, Posterior & lateral fascial compartment,
each having its own muscles, nerves, and arteries.

The interosseous membrane:
A strong membrane attaches to interosseous borders of ‘Radius &‘Ulna





The anterior compartment (Flexors)

- Muscles:

Superficial group:

Palmaris longus,
Pronator teres,
Flexor carpi radialis,
Flexor carpi ulnaris;

Intermediate group:

Flexor digitorum superficialis

Deep group:

Flexor pollicis longus,
Flexor digitorum profundus,
Pronator quadratus

Superficial – intermediate groups of muscles: Have a common tendon of origin, attached to the Medial epicondyle of 'humerus.

Nerve Supply
ALL – 1 ½

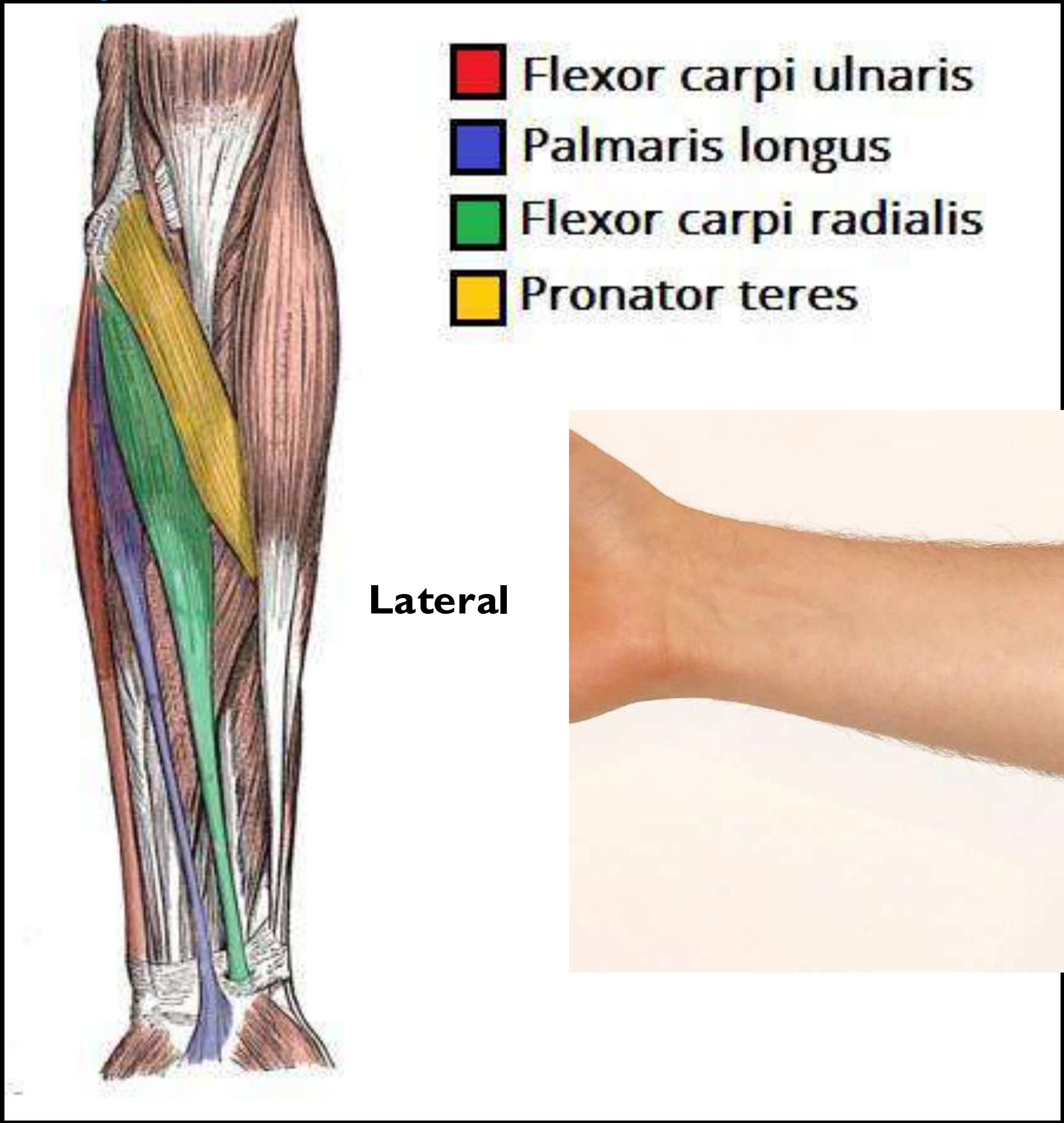
- Blood supply: Ulnar & Radial arteries

- Nerve supply:

All the muscles → by median nerve and its branches,
Except flexor carpi ulnaris & medial part of flexor digitorum profundus→ by ulnar nerve.

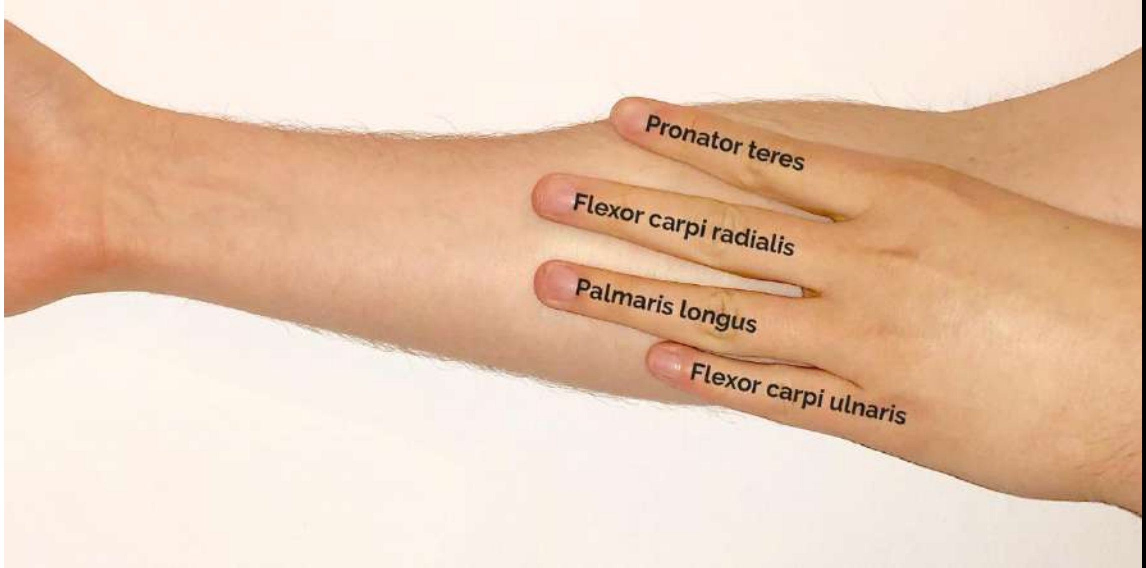
THE ANTERIOR COMPARTMENT (FLEXORS)

Medial epicondyle



Lateral

Superficial group:

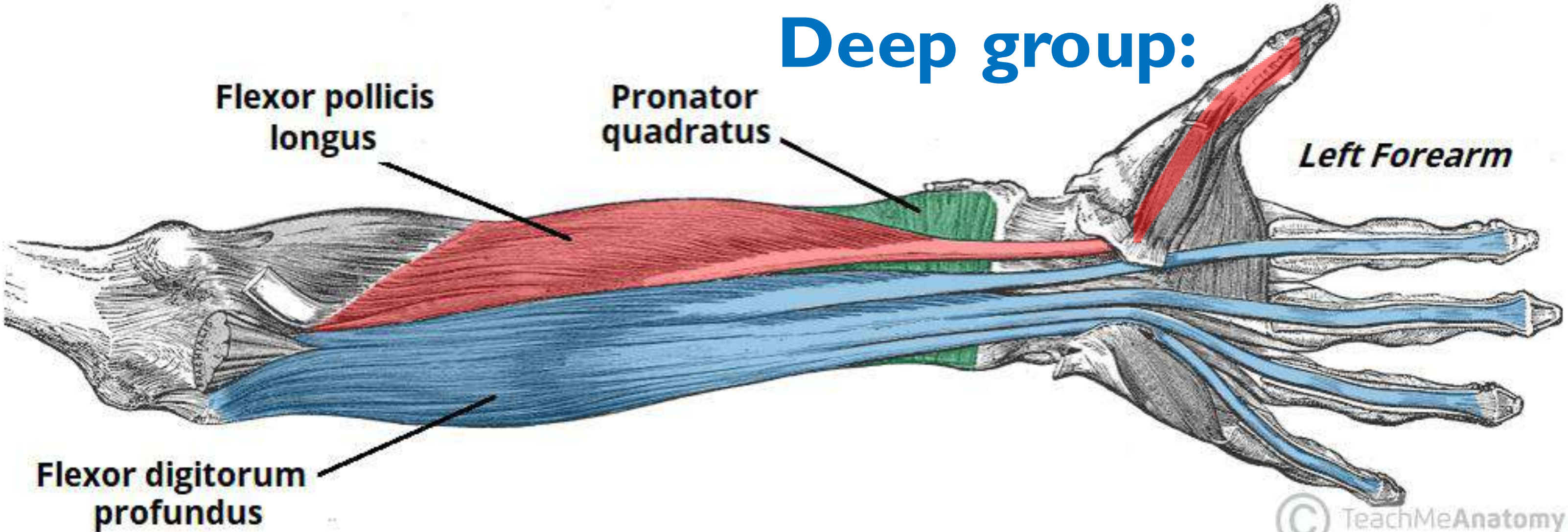


THE ANTERIOR COMPARTMENT (FLEXORS)

Intermediate group:

Flexor digitorum superficialis

Deep group:



Anterior compartment Superficial group

Pronator teres

- Origin:

Humeral head: Medial epicondyle of humerus

Ulnar head: Medial border of coronoid process of ulna

- Insertion: middle of Lateral aspect of shaft of Radius

- Nerve supply:

Median Nerve

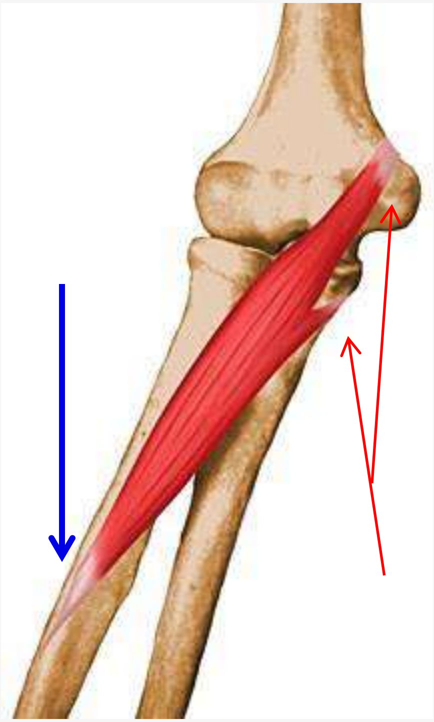
- Action:

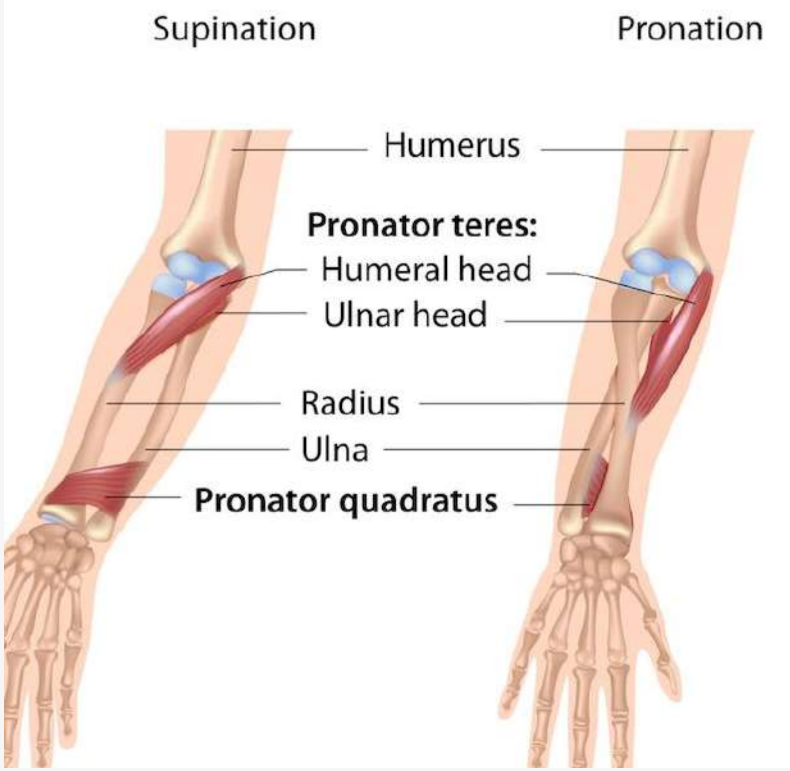
- Pronation

& flexion of forearm.

Supination / pronation:

https://www.youtube.com/shorts/zlQLHzHQCMc





Anterior compartment Superficial group

Flexor Carpi Radialis

-Origin:

Medial epicondyle of humerus

- Insertion: Bases of 2nd ± 3rd metacarpals

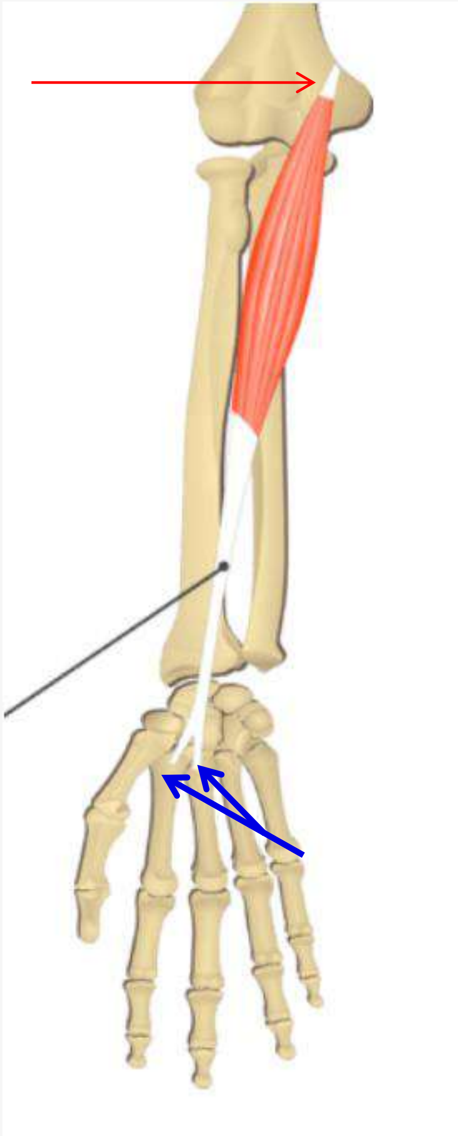
- Nerve supply:

Median Nerve

- Action:

- Flexion &

Abduction (radial / lateral deviation) of 'hand at wrist joint.



Anterior compartment Superficial group

Flexor Carpi Ulnaris

-Origin:

Humeral head: Medial epicondyle of humerus

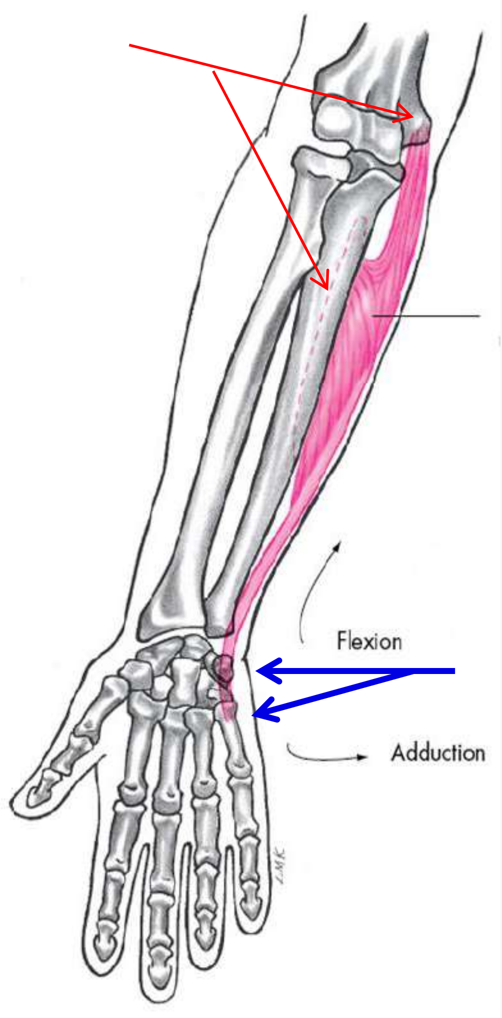
Ulnar head: Medial aspect of olecranon process
& posterior border of ulna

-Insertion: Pisiform & Hamate Carpal bones
& base 5th metacarpal

- Nerve supply:
  Ulnar Nerve

- Action:
  - **Flexion**

& **Adduction** (ulnar / medial deviation)
of hand at wrist joint.



Anterior compartment Superficial group

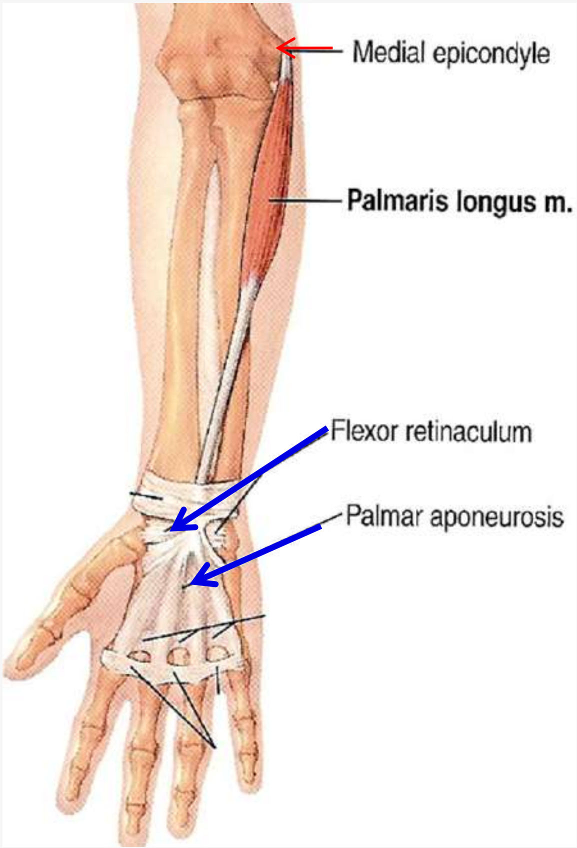
Palmaris longus

- Origin:
  Medial epicondyle of humerus

- Insertion: Flexor retinaculum & palmar aponeurosis

- Nerve supply:
  Median Nerve

- Action:
  - weak flexion of hand .



Palmaris longus:
Sometimes present/ absent.
- clinically:
Its long tendon is used in tendon repair surgery.

Anterior compartment Intermediate group

Flexor Digitorum Superficialis

- Origin:

Humero-ulnar head: Medial epicondyle of humerus;
medial border of coronoid process of ulna

Radial head: Oblique line on anterior surface of shaft of radius

- Insertion:

Middle phalanx of medial four fingers

- Nerve supply:

Median Nerve

- Action:
  - Flexion of middle phalanx of medial 4 fingers &
  - Helps in flexion of hand at wrist joint

Anterior compartment Deep group

Flexor Digitorum Profundus

- Origin:

  Upper ¾ Anteromedial surface of shaft of

  Ulna + IOM

-Insertion:

  Distal phalanges of 'medial four fingers

- Nerve supply:

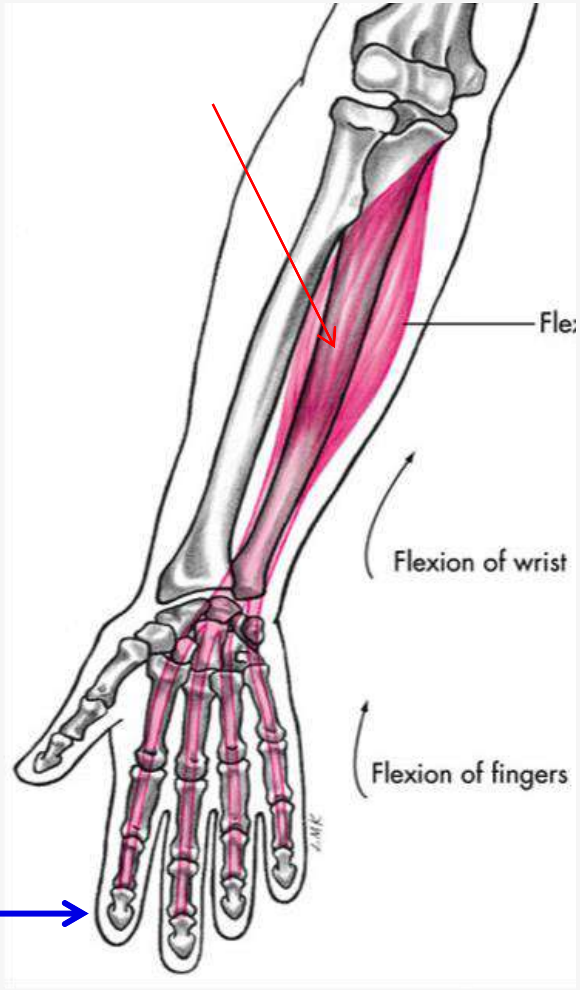
  Ulnar nerve (medial half)

  Median nerve (lateral half)

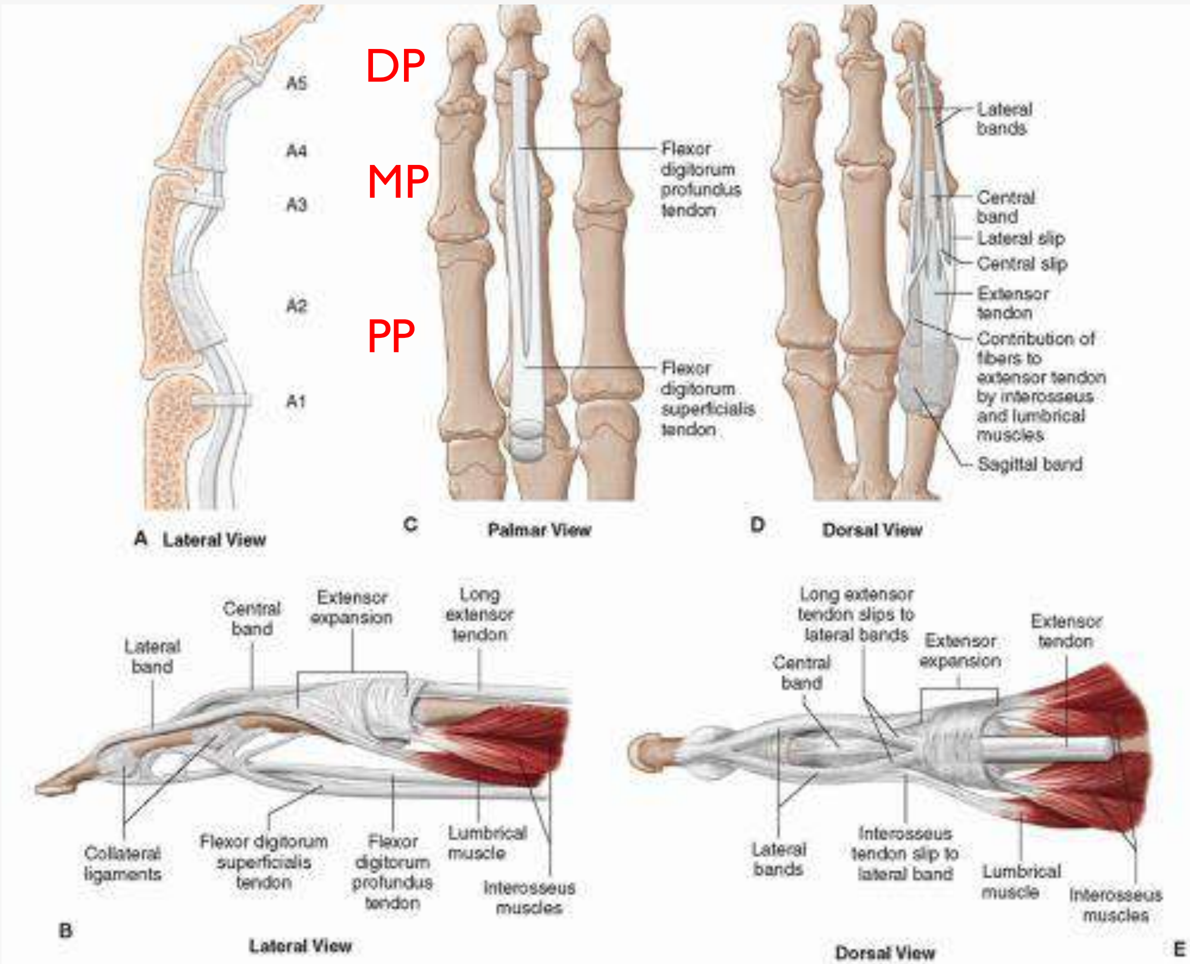
  (anterior interosseous br.)

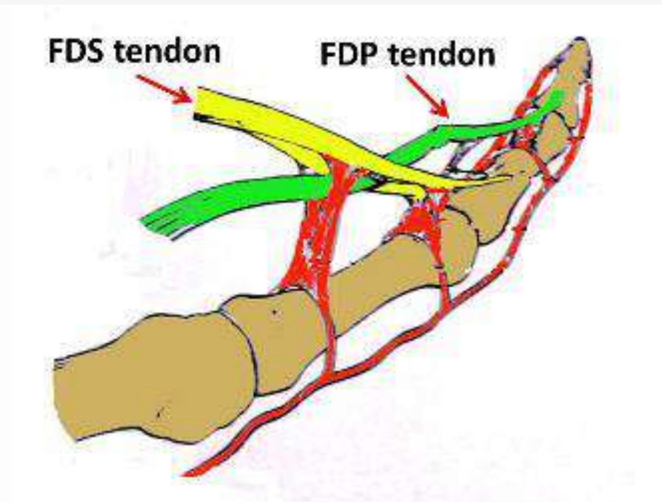
- Action:

  - Flexes distal phalanx of 'fingers
  - Assists in flexion of middle & proximal phalanges
  - Helps in flexion of hand at wrist joint



DIFFERENCE IN THE INSERTION OF: FLEXOR DIGITORUM SUPERFICIALIS & PROFUNDUS AND EXTENSOR DIGITORUM





The tendon of FDS is inserted into:

- the front of MP

Anterior compartment Deep group

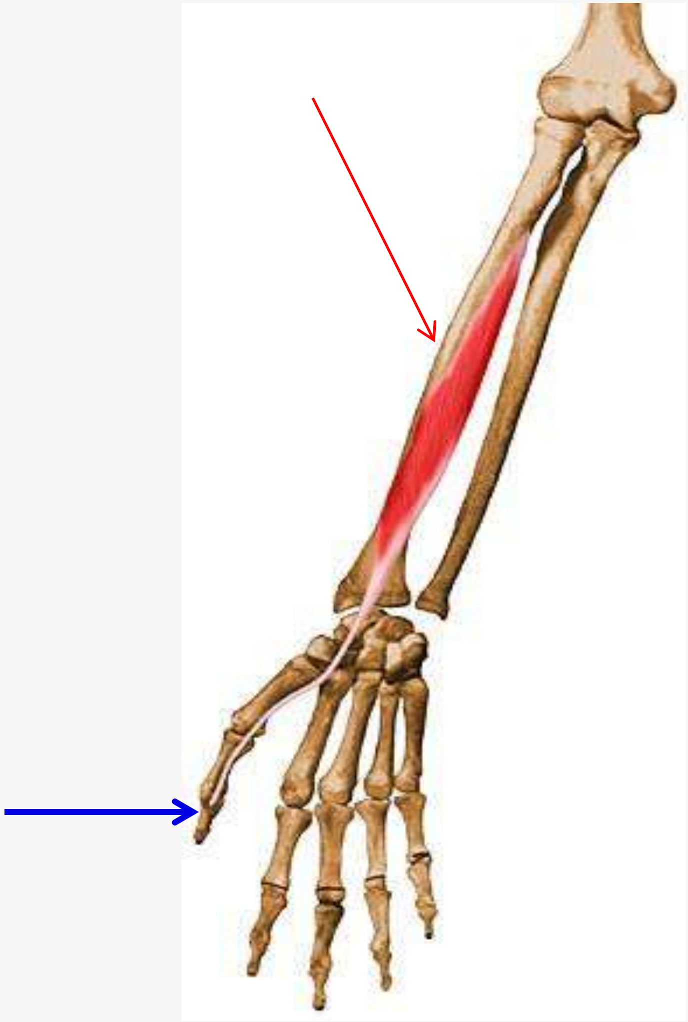
Flexor Pollicis Longus

- Origin:
  middle of Anterior surface of shaft of **Radius + IOM**

-Insertion: Distal phalanx of ‘**thumb**

- Nerve supply:
  Median Nerve
  (anterior interosseous br.)

- Action:
  Flexes distal phalanx of ‘thumb



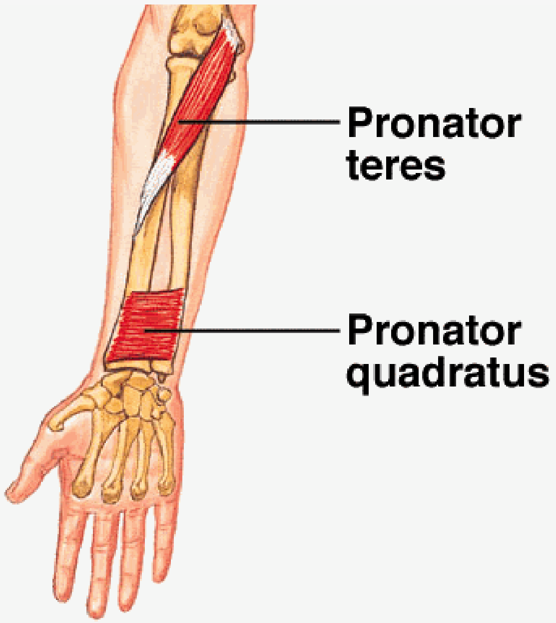
Anterior compartment Deep group

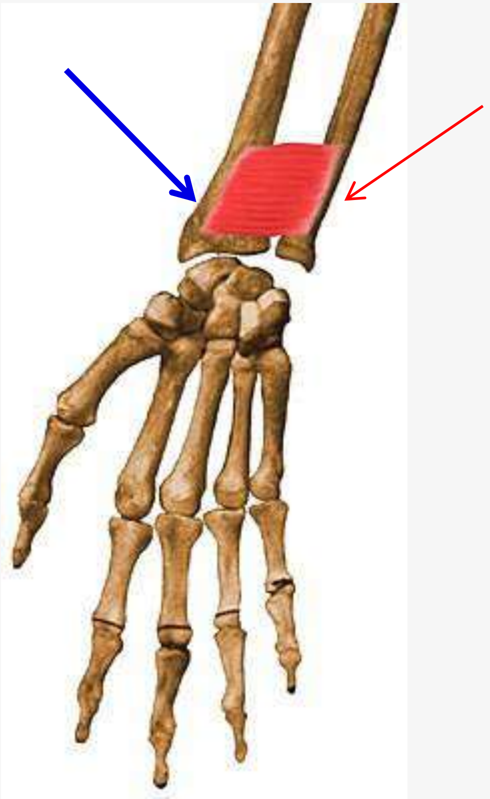
Pronator Quadratus

- Origin:
  Lower 1/4 of Anterior surface of shaft of **Ulna**
- Insertion:
  Lower 1/4 of Anterior surface of shaft of **Radius**

- Nerve supply:
  Median Nerve
  (anterior interosseous br.)

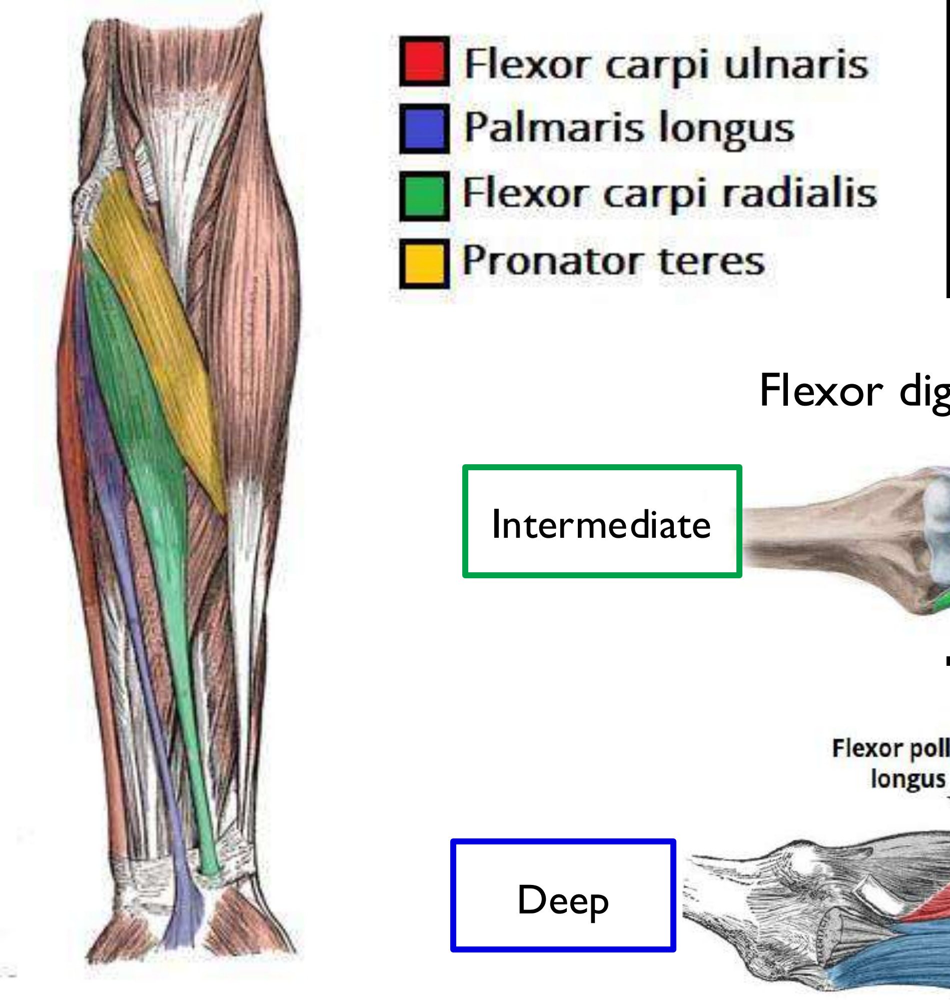
- Action:
  Pronates the forearm





THE ANTERIOR COMPARTMENTS (FLEXORS)

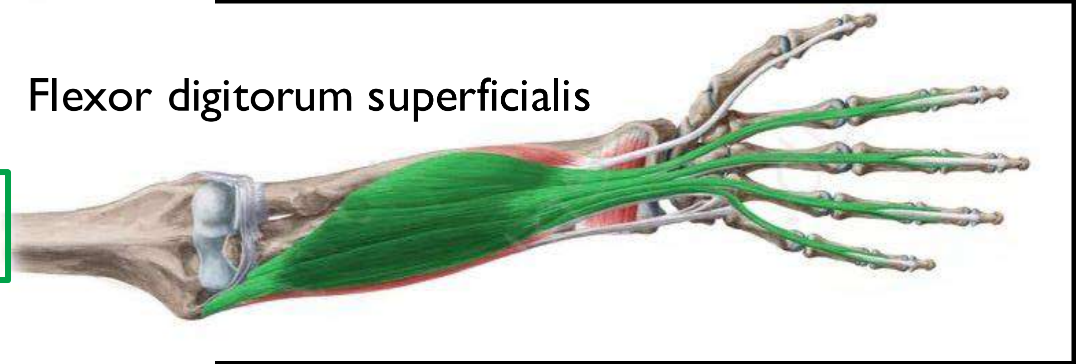
Superficial



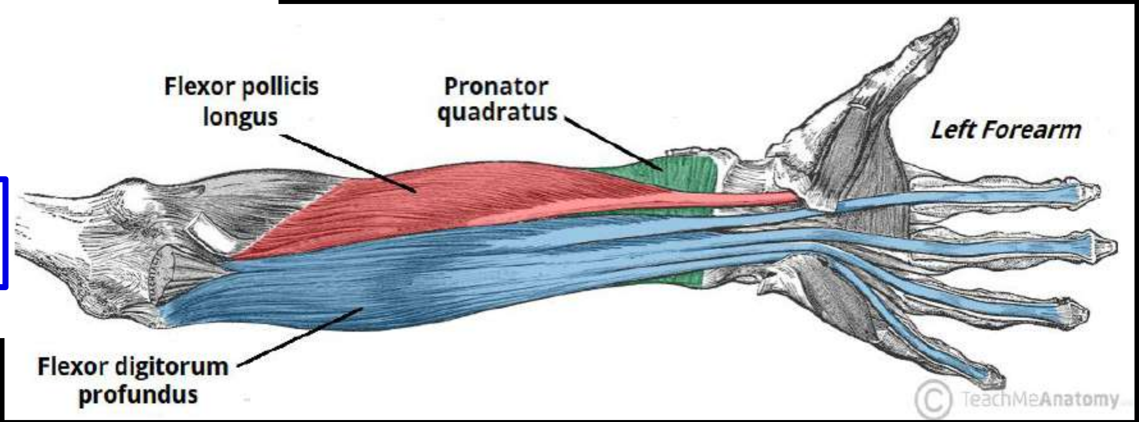
Nerve Supply
ALL – 1 ½

Flexor digitorum superficialis

Intermediate

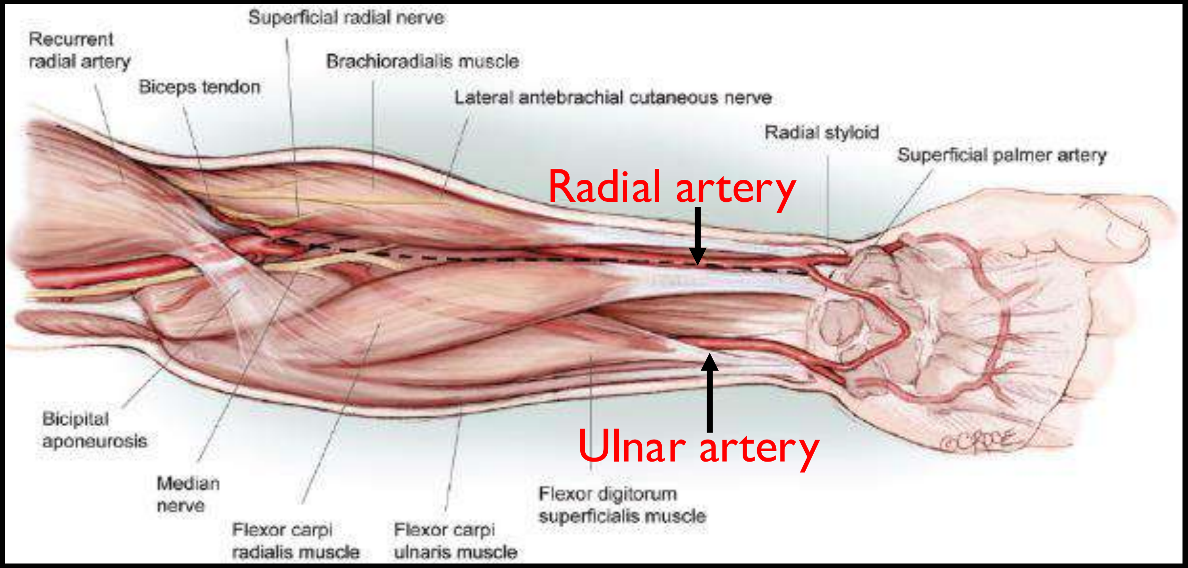


Deep

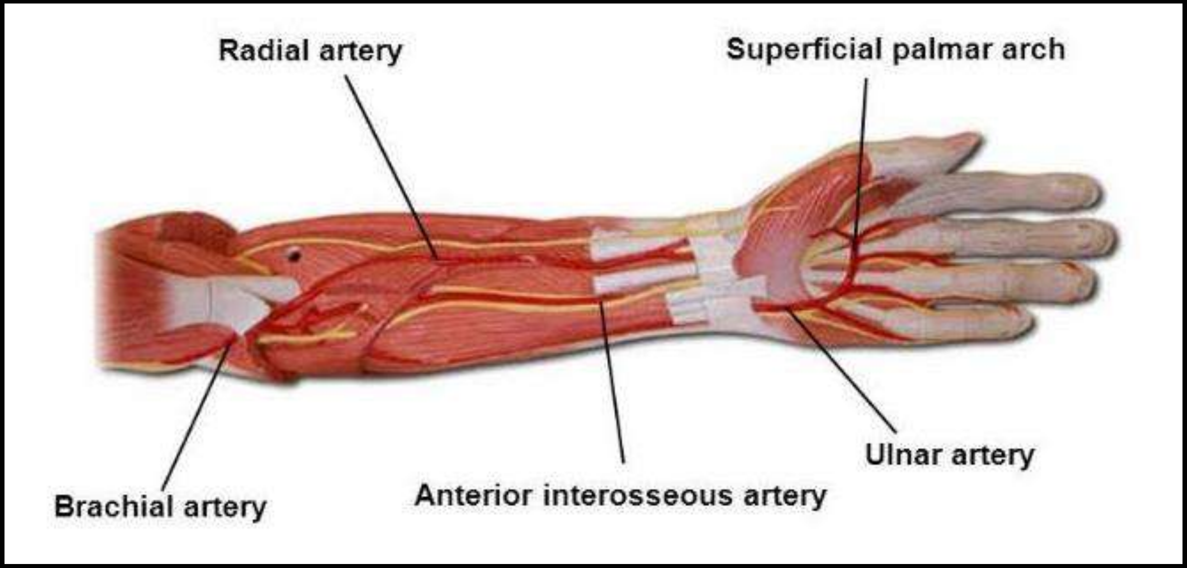


The Arteries of the Forearm

Ulnar artery
Radial artery



Begins
Ends
Branches
Relations
Site of pulsation



The Ulnar Artery

The **larger** of the two terminal branches of **brachial artery**

- Begins: In the cubital fossa at the level of the neck of ‘radius.

It descends through the anterior compartment of the forearm

It enters ‘palm in front / Superficial to ‘flexor retinaculum

lateral to the ‘ulnar nerve

- Ends: by forming the superficial palmar arch

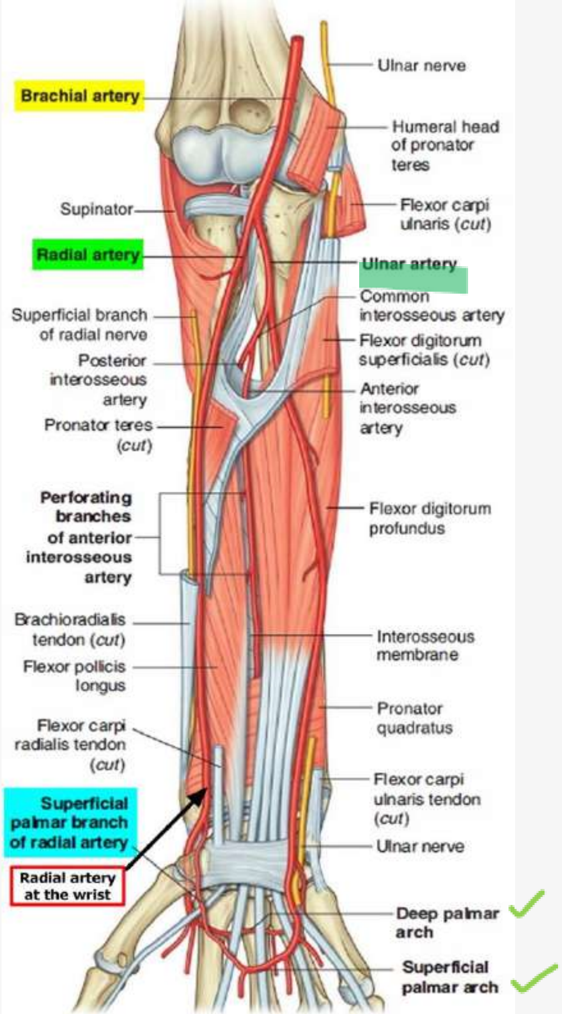
-Relations:

-Its upper part:

lies **deep** to the muscles.

-Its lower part:

it becomes **superficial & lateral** to ulnar nerve between tendons of FCU & FDS



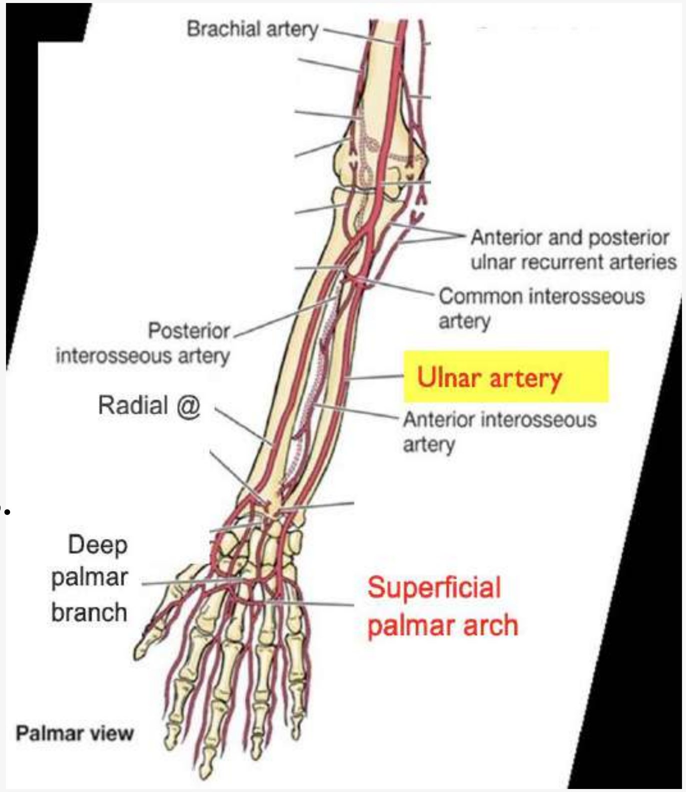
The Ulnar artery

- Branches:

■ Muscular branches

■ **anterior & posterior** Ulnar Recurrent branches:
anastomosis around **elbow** joint.

■ The **common** interosseous artery:
arises from the upper part of the ulnar artery and divides into: the anterior & posterior interosseous arteries
(in front & behind IOM)
They provide nutrient arteries to the **radius & ulna bones**.



■ anastomosis branches around the **wrist** joint.

■ the Deep palmar branch of ulnar artery: it joins the radial artery.

Site for taking Ulnar pulse:
-Front of 'flexor retinaculum,
lateral to pisiform bone
covered only by skin & fascia.

The Radial Artery

It is the **smaller** of the two terminal branches of **brachial artery**

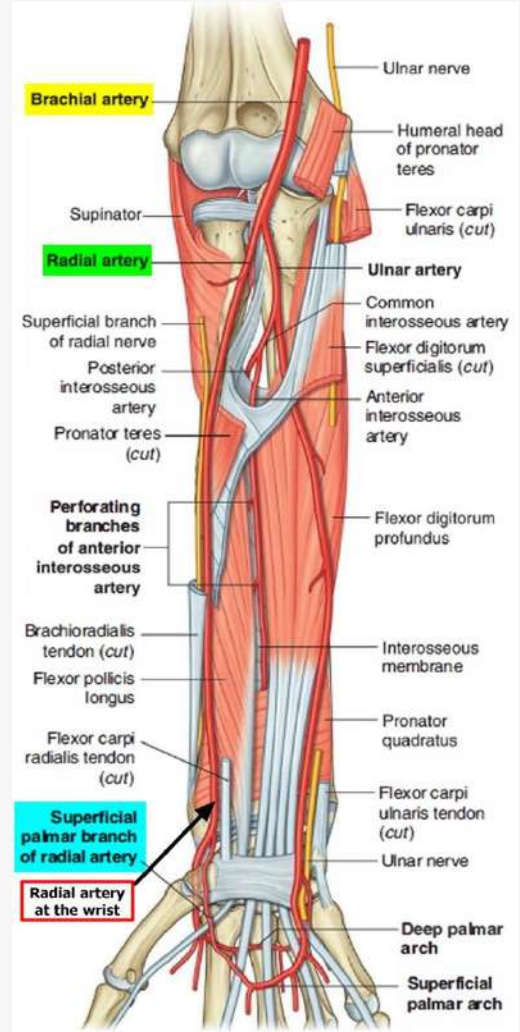
- Begins: In the cubital fossa at the level of the neck of 'radius.

The radial artery lies medial to 'superficial branch of the radial nerve.
Lateral to tendon of FCR.
It winds around the wrist to reach the back of the hand, passing through 'anatomical snuffbox.

- **Ends**: by forming the deep palmar arch

- **Relations:**

it is between tendons of brachioradialis laterally & flexor carpi radialis medially.



The Radial artery

- Branches:

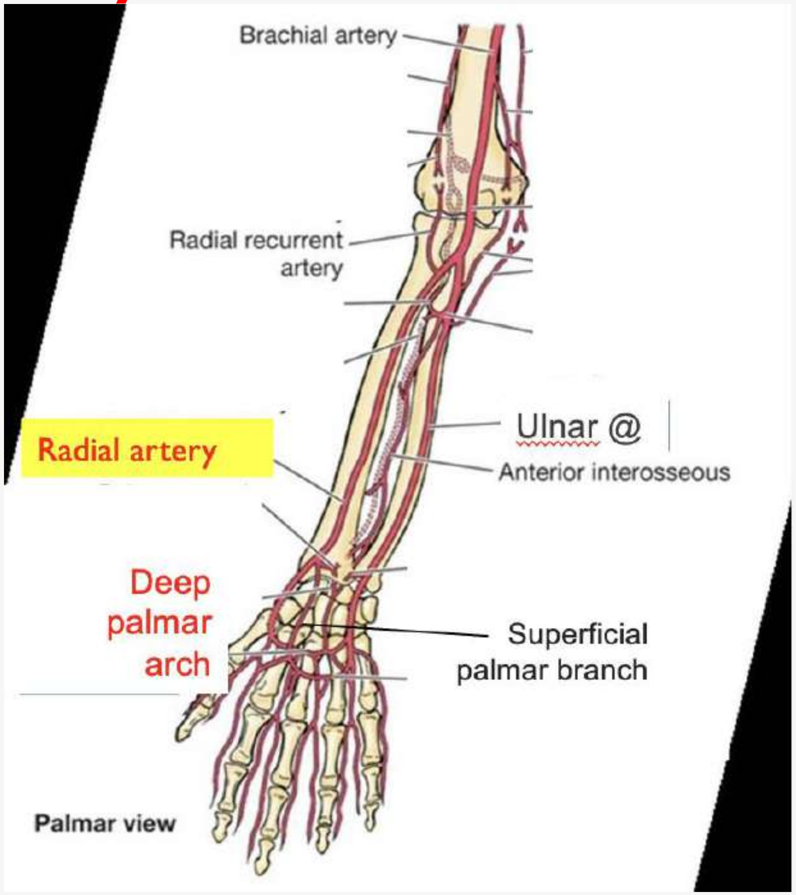
■ Muscular branches

■ Radial Recurrent branches:

anastomosis around the **elbow** joint.

■ The superficial palmar branch of radial artery:

joins the ulnar artery.



Site for taking Radial pulse:

In the distal part of forearm,

on anterior surface of radius

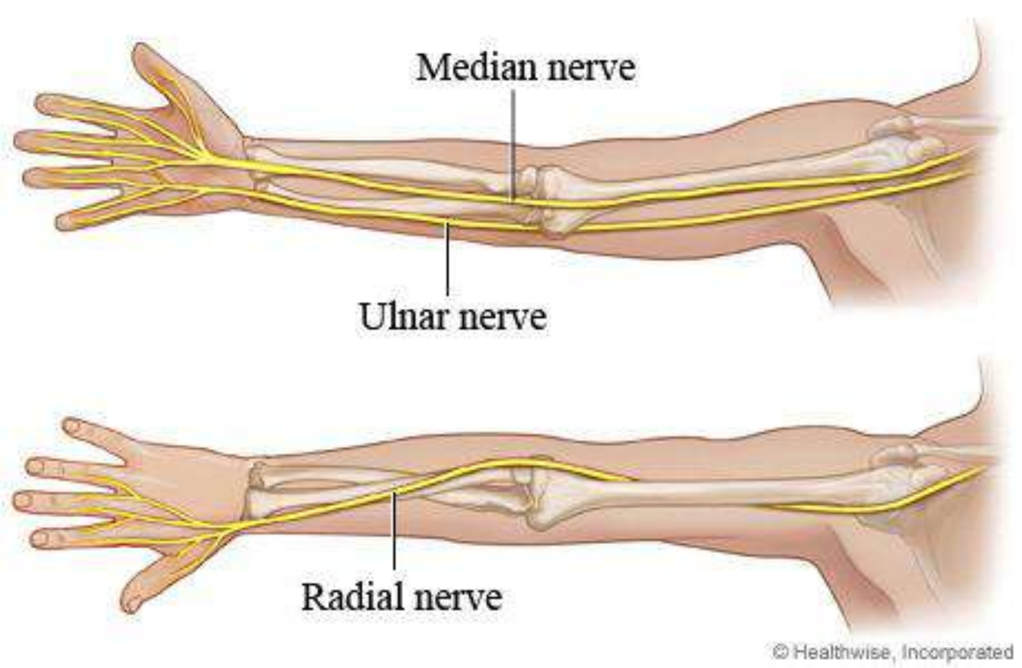
covered only by skin & fascia

OR Within the anatomical snuffbox.

The Nerves of the Forearm

Median nerve
Ulnar nerve
Radial nerve

Begins
Ends
Branches
Relations



Ulnar Nerve ~ In the forearm

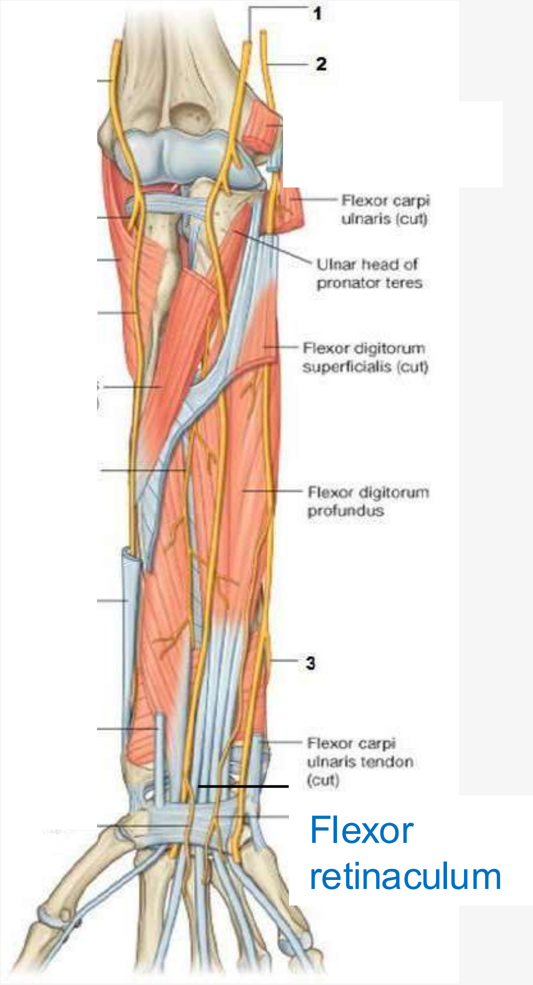
~It passes **behind** 'medial epicondyle of humerus

~It enters the front of 'forearm
by passing between the two heads of '**FCU**.
~It descends deep between 'FCU & FDP.

~In the distal forearm:
the ulnar nerve is **medial** to the ulnar artery.

~ At the wrist:
the nerve becomes <u>superficial</u> &
lies between tendons of FCU & FDS.

~It enters the palm by passing
in front (superficial to) of the **flexor retinaculum**
& lateral to the **pisiform bone**
& **medial** to the ulnar artery



Branches in forearm:

- **Muscular branches:**
  To FCU & medial half of FDP & in the hand ….

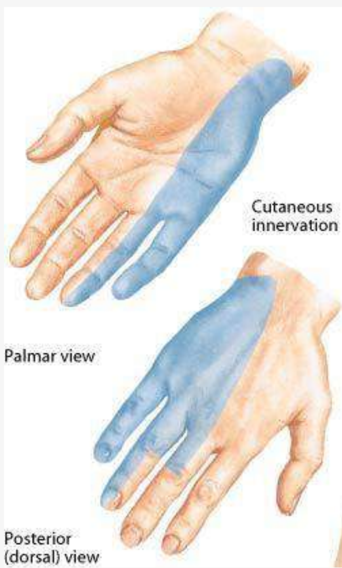
- **Articular branches:** to the elbow & wrist joint

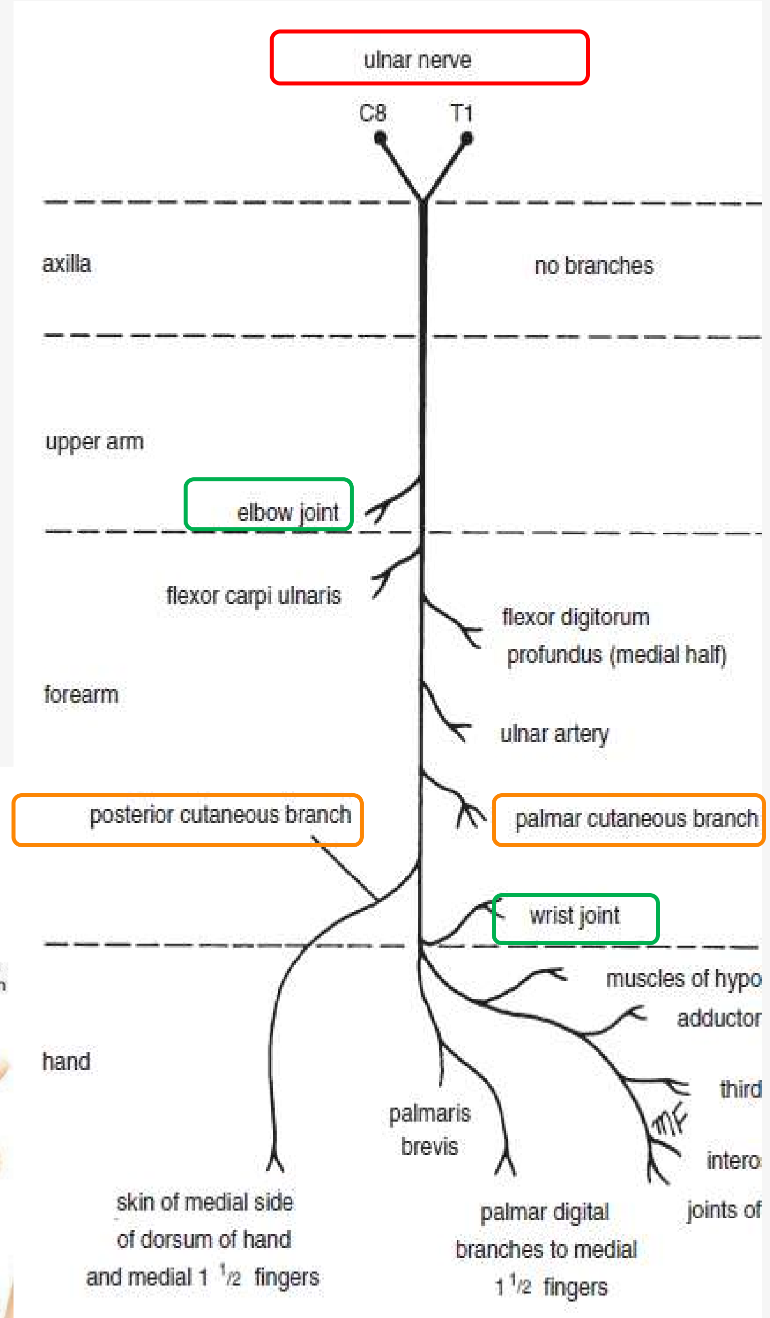
- **Palmar cutaneous branch:**

a small branch
arises in the middle of ‘forearm
It supplies skin over ‘hypothenar eminence. & …

- **Dorsal (posterior) cutaneous branch:**

a large branch
arises in the lower of ‘forearm.
It supplies skin on ‘posterior surface of ‘hand & fingers.





Median Nerve ~ In the forearm

Course:

- It leaves the cubital fossa to enter the forearm by passin
  between the two heads of the **pronator teres**.
- It continues downward between the FDS & the FDP.

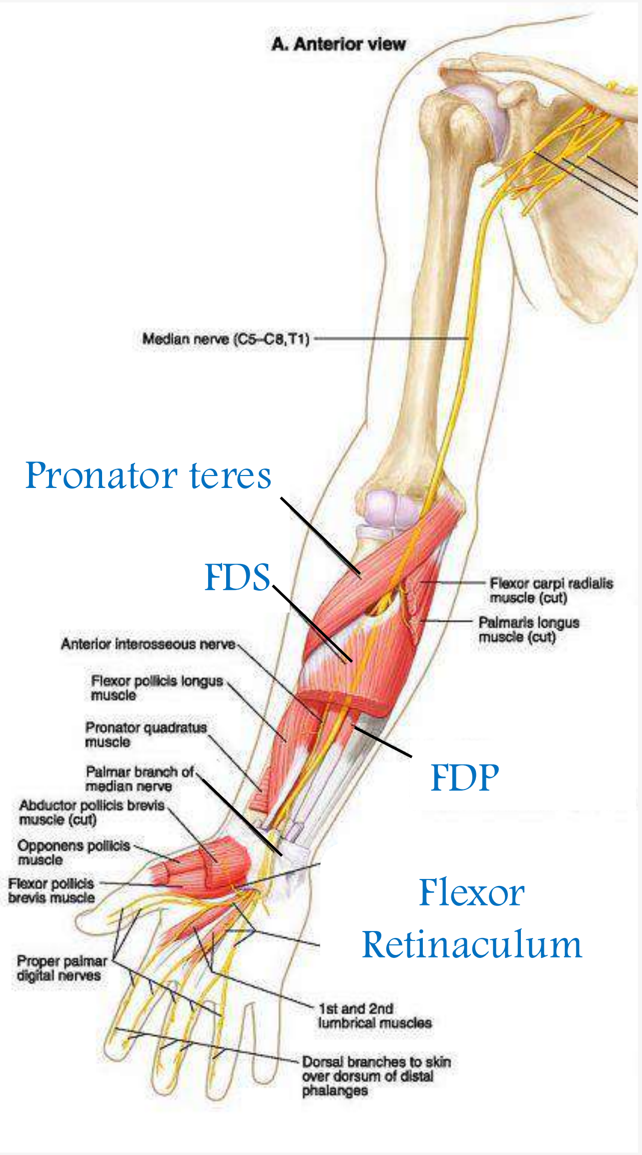
At the wrist:

it emerges lateral to the FDS &
lies behind 'tendon of palmaris longus.

It enters the palm by passing through the carpal tunnel behind (deep to) the flexor retinaculum.

~**Ends**:

In the palm as the palmar cutaneous branch.



Branches in forearm:

■ **Muscular branches:**

to forearm muscles: all …except …

and hand muscles (thenar ms & 1^{st}~2^{nd} lumbricals)

■ **Articular branches:** to the **Elbow** joint

■ **Anterior interosseous nerve**

~ It arises from 'nerve as it emerges between two heads of pronator teres.

~It descends with ' anterior Interosseous artery on 'front of 'IOM, between FPL & FDP.

~ It ends on the anterior surface of 'carpus

Branches

■ Muscular branches:

FPL, PQ, the lateral half of **FDP**.

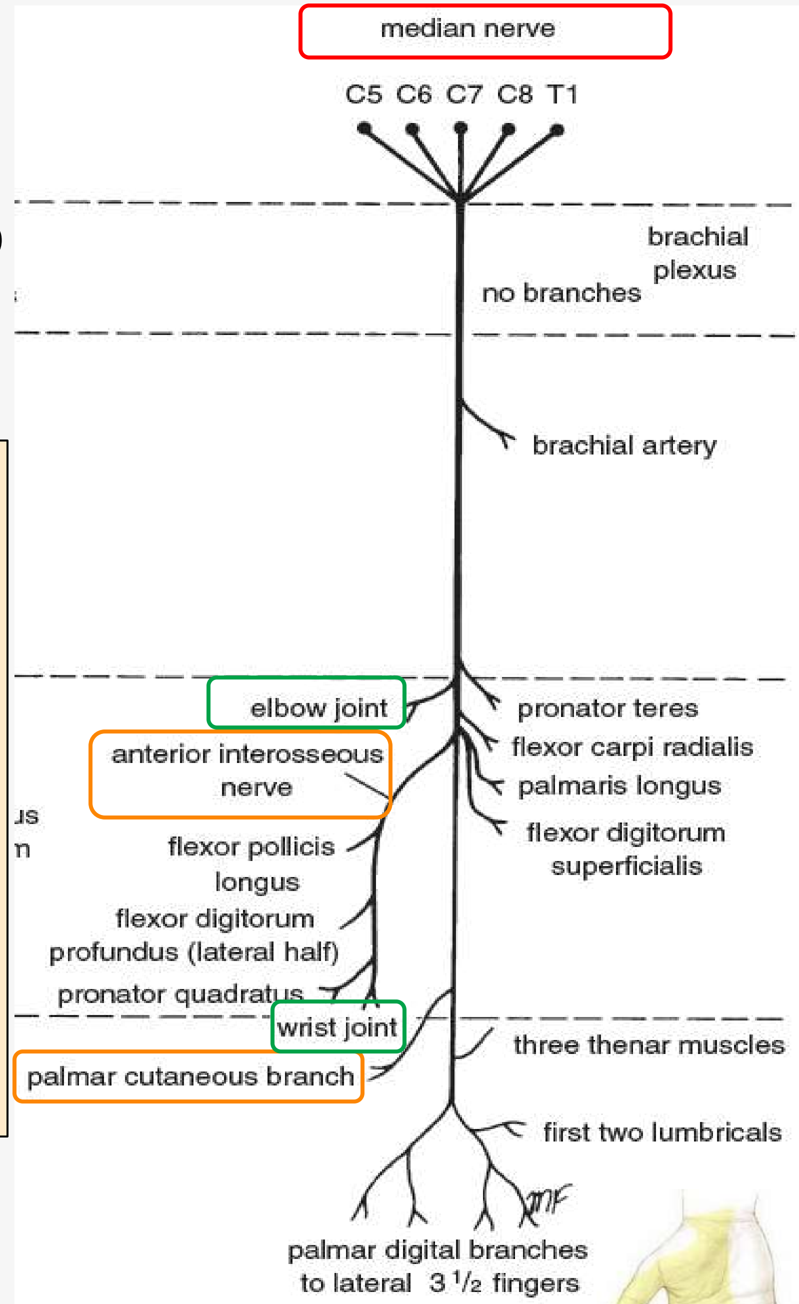
■ Articular branches:

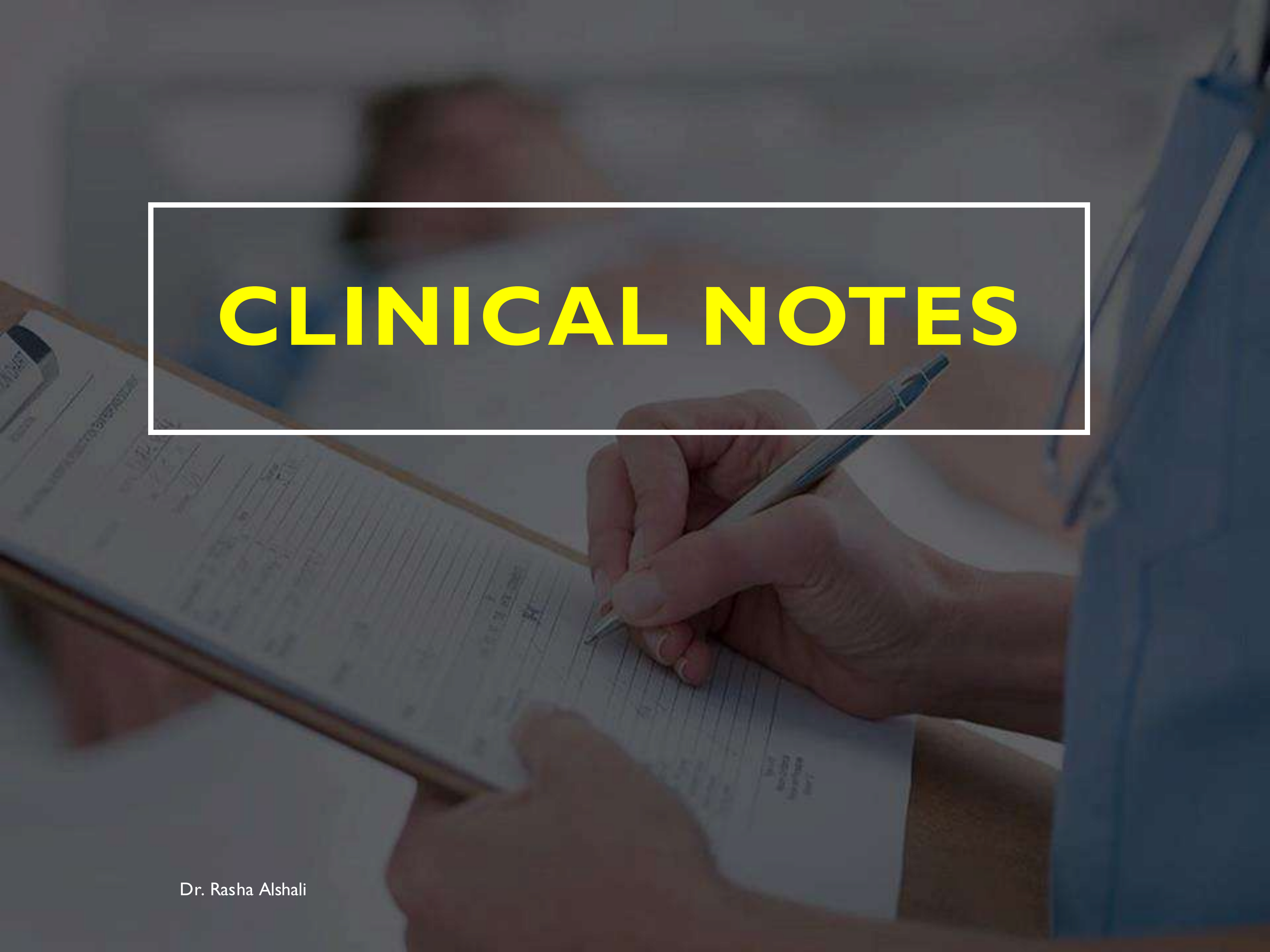
wrist & inferior radioulnar J & joints of the hand.

■ **Palmar cutaneous branch:**

Arises from the median n. before passing behind flexor retinaculum,

supplies skin over 'lateral 2/3 of the palm, and lateral 3½ fingers.





CLINICAL NOTES

Dr. Rasha Alshali

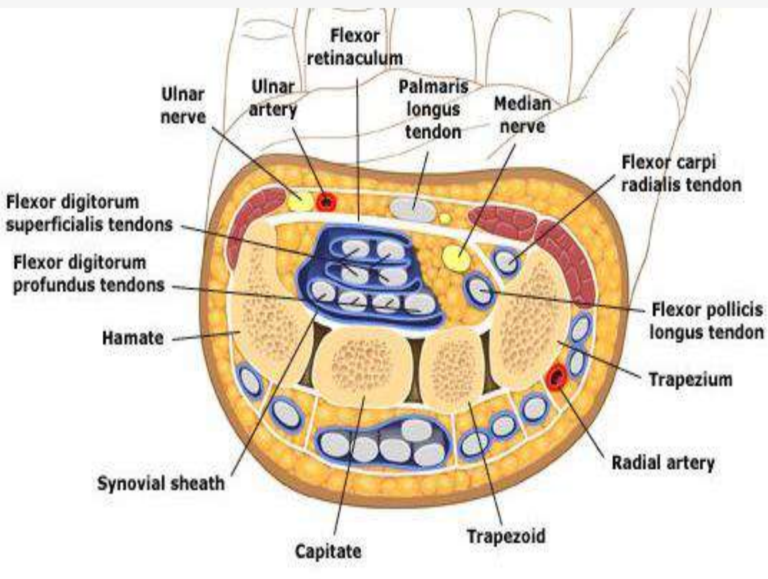
The Carpal Tunnel

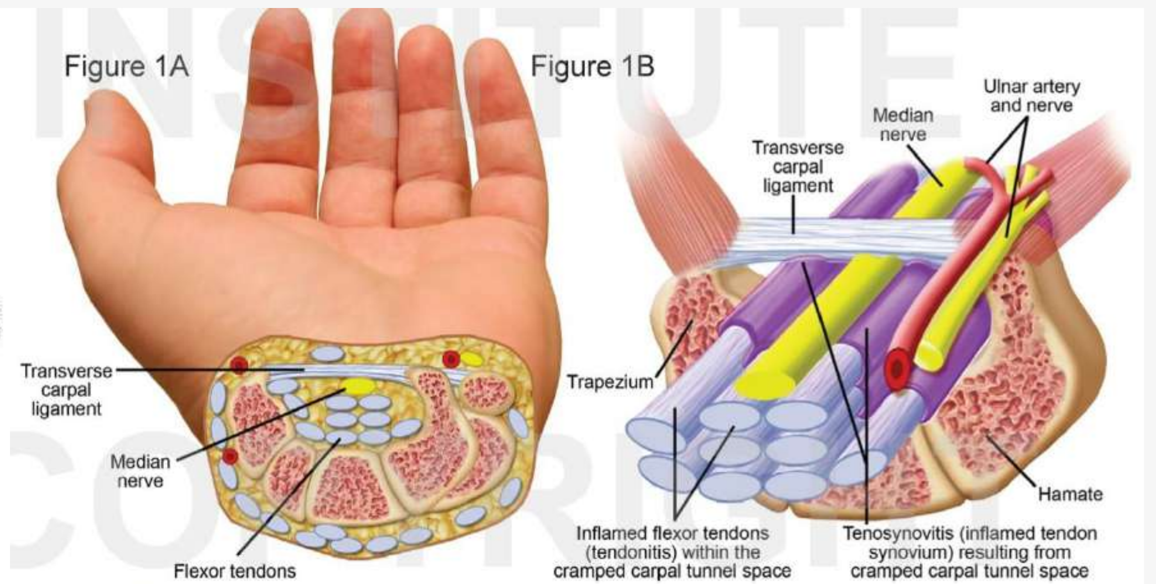
How is it formed??

The carpal tunnel is formed by the concave anterior surface of the carpal bones and closed by the flexor retinaculum,

The carpal tunnel is for the passage of:

- the **median nerve**
- the flexor tendons of the thumb and fingers.



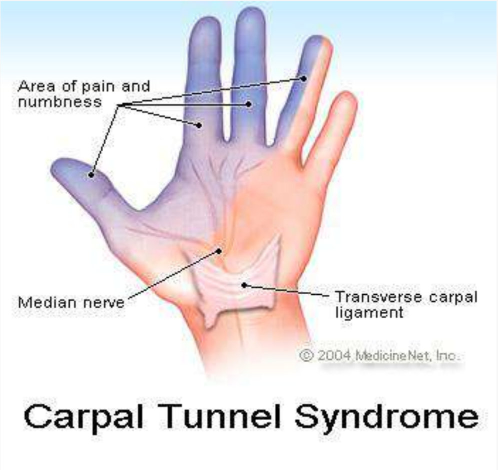


Carpal Tunnel Syndrome (C T S)

CTS **is a medical condition.** It is the compression of the Median nerve in the restricted space between the tendons through the carpal tunnel at the wrist.

Cause: compressed median nerve is due to

- Edema to the flexor tendons
- Fracture or dislocation of the carpal bones
- Fibrosis or thickening of the flexor retinaculum



Symptoms:

Motor affection:

Paralysis of the thenar muscles & the 1^{st},2^{nd} Lumbrical muscles

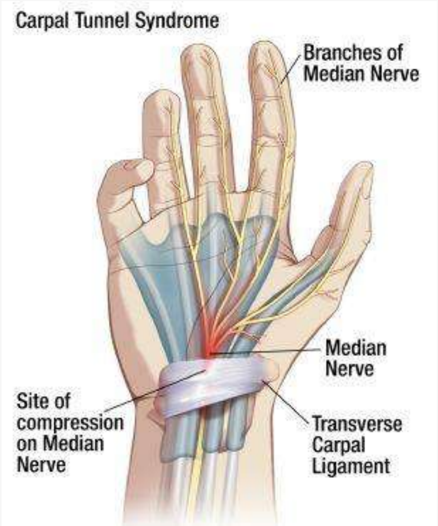
Sensory affection:

Sever pain

burning pain & numbness along the distribution of ‘nerve.

No Sensory loss !! Why

as the cutaneous branch of median nerve passes **superficial** to the flexor retinaculum.



The management:

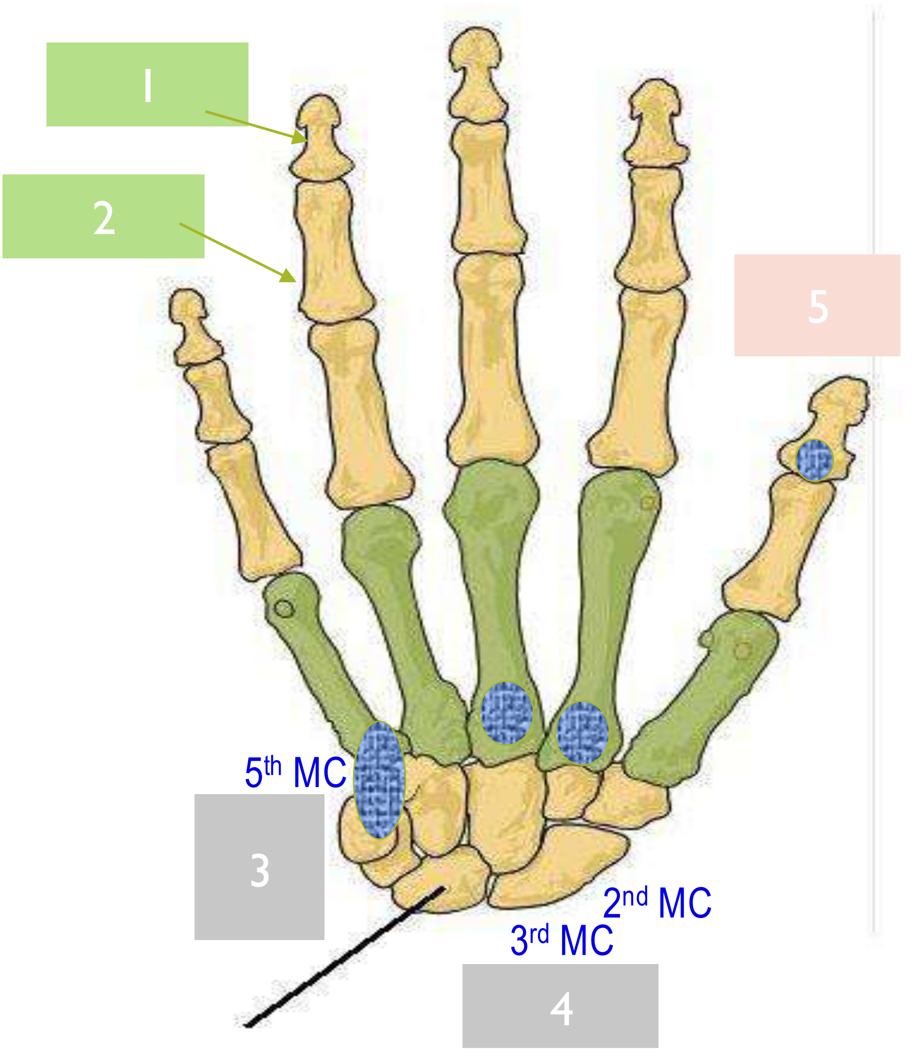
Decompressing the tunnel by making a longitudinal incision through ‘flexor retinaculum

TO BE CONTINUED


HOME
WORK

Insertions of long tendons in 'hand bones

FRONT of the hand



The Posterior compartment (Extensors)

Superficial group of muscles:
Have a common tendon of origin, attached to:
Lateral epicondyle of the humerus.

Contents:

- Muscles:

Superficial group:

Extensor carpi radialis brevis,
Extensor carpi ulnaris,
Extensor digitorum,
Extensor digiti minimi,
Anconeus.

Deep group:

Extensor indicis,
Extensor pollicis brevis,
Extensor pollicis longus,
Abductor pollicis longus,
Supinator

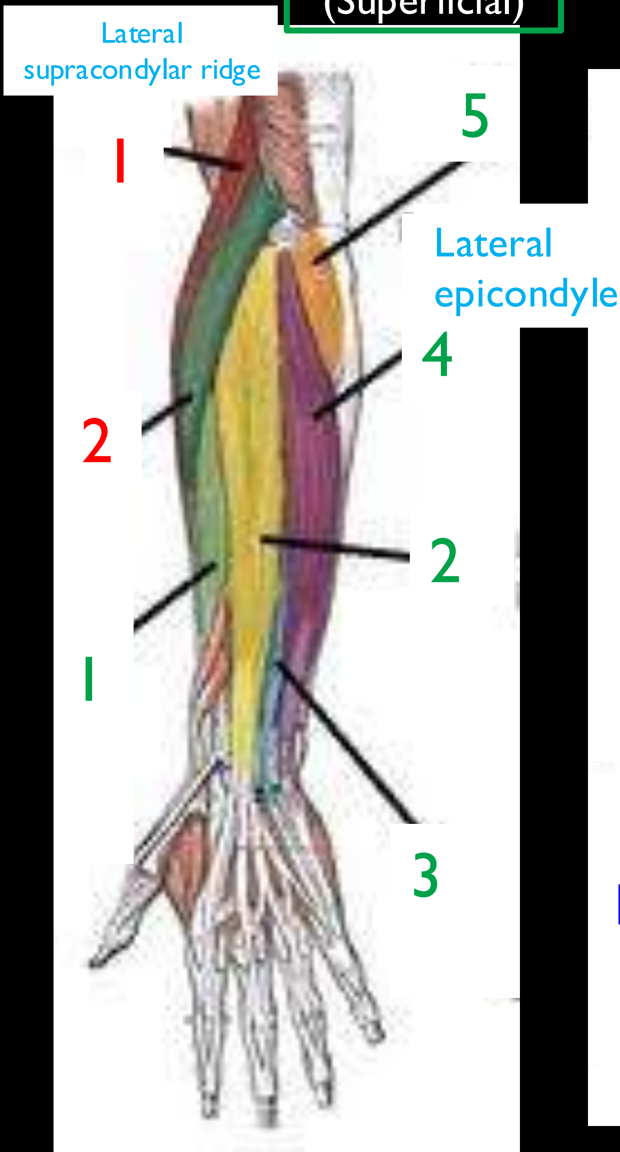
- Blood supply: Posterior & Anterior interosseous arteries (Ulnar artery)
- Nerve supply: Deep branch of the Radial nerve

Except Anconeus → by Radial nerve

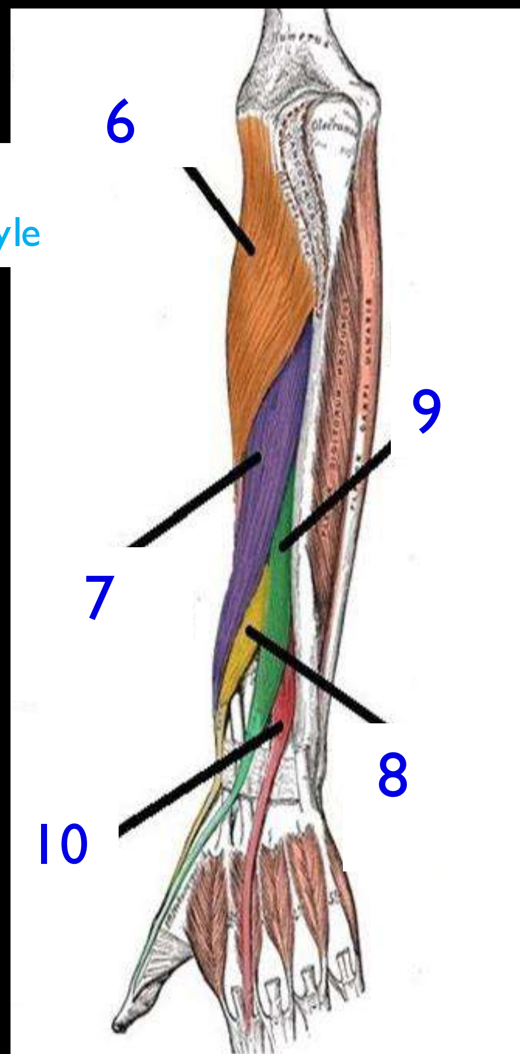
THE POSTERIOR – LATERAL COMPARTMENTS (EXTENSORS)

Lateral

Posterior (Superficial)



Posterior (Deep)



Nerve Supply
ALL – 3

1- Brachioradialis
2- Extensor carpi radialis longus

1. Extensor carpi radialis brevis
2. Extensor digitorum
3. Extensor digiti minimi
4. Extensor carpi ulnaris
5. Anconeus

6. Supinator
7. Abductor pollicis longus
8. Extensor pollicis brevis
9. Extensor pollicis longus
10. Extensor indicis

The Lateral compartment

The lateral fascial compartment may be considered a part of posterior compartment.

Two muscles have a common origin:
Lateral supracondylar ridge of humerus.

Contents:

- Muscles: 1- Brachioradialis
  2- Extensor carpi radialis longus
- Blood supply: Radial artery
  Brachial artery
- Nerve supply:

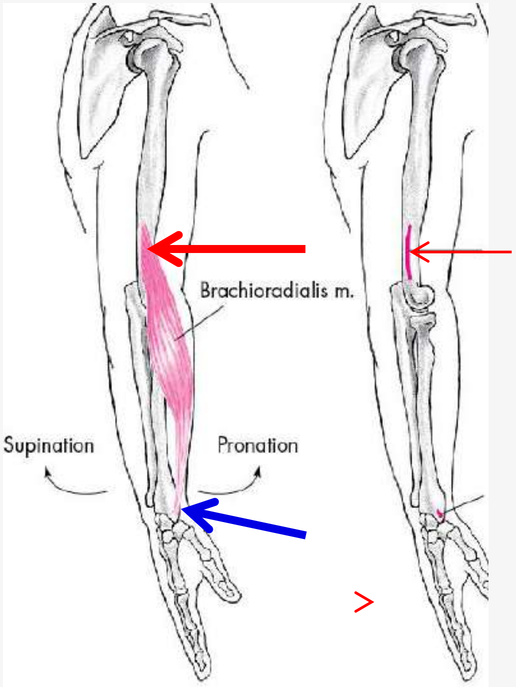
Radial nerve

Nerve
Supply

ALL

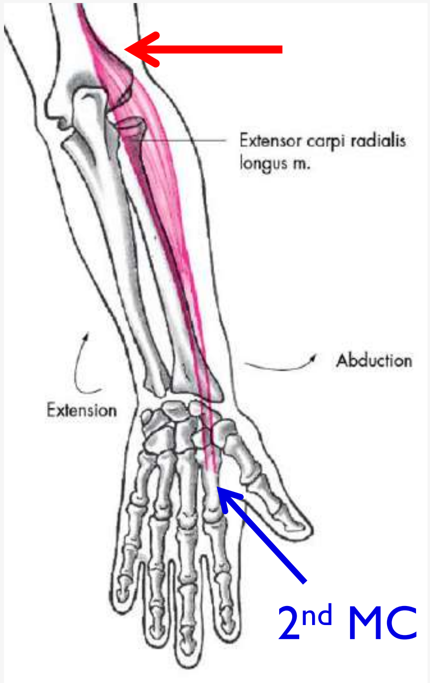
Brachioradialis

- Origin:
  Lateral supracondylar ridge of **humerus**
- Insertion:
  Base of styloid process of **radius**
- Action:
  Flexion of elbow joint;
  Rotates forearm to midprone position.



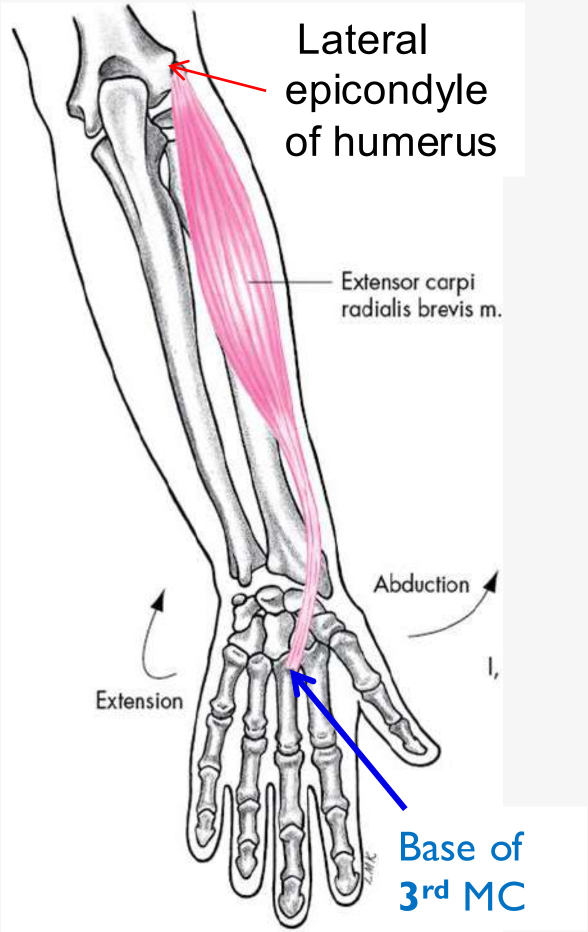
Extensor carpi radialis longus

- Origin:
  Lateral supracondylar ridge of **humerus**
- Insertion:
  Posterior surface of base of 2nd metacarpal bone
- Action:
  Extension &
  Abduction (radial / lateral deviation) of ‘hand at wrist joint.



Posterior compartment Superficial group

Extensor carpi radialis brevis



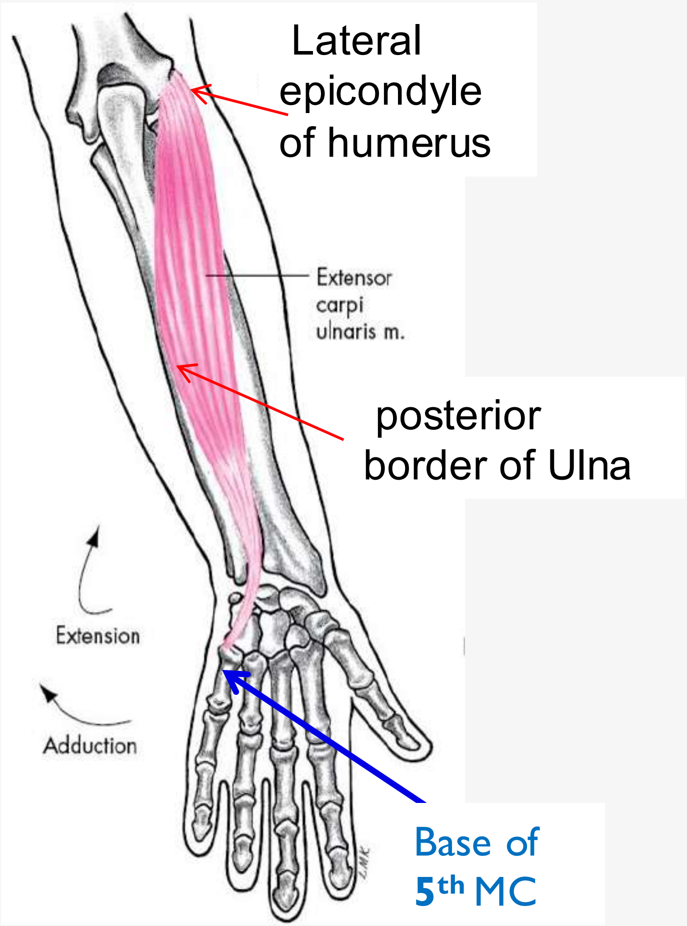
Action:

- Extension &

abduction (lateral deviation)

of hand at wrist joint

Extensor carpi ulnaris



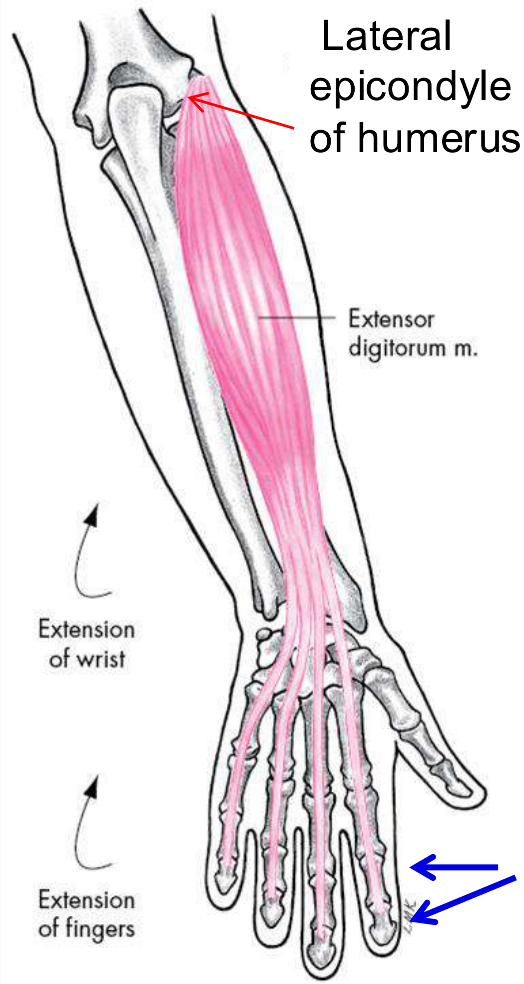
Action:

- Extension &

adduction (medial deviation)

of hand at wrist joint

Extensor digitorum

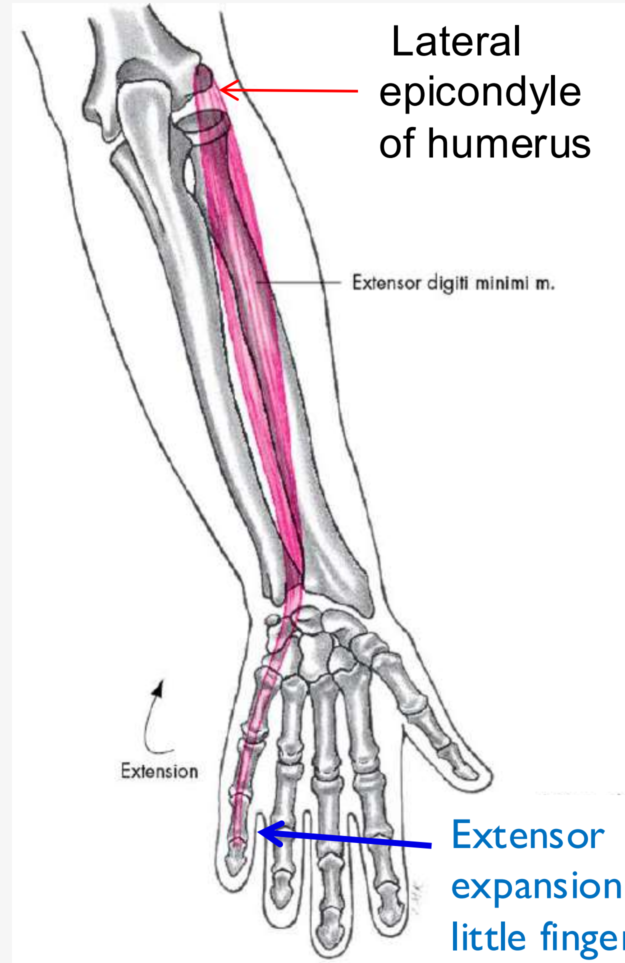


(Extensor expansion) **Middle & distal phalanges** of medial four fingers

Action:

- Extension of fingers & hand

Extensor digiti minimi

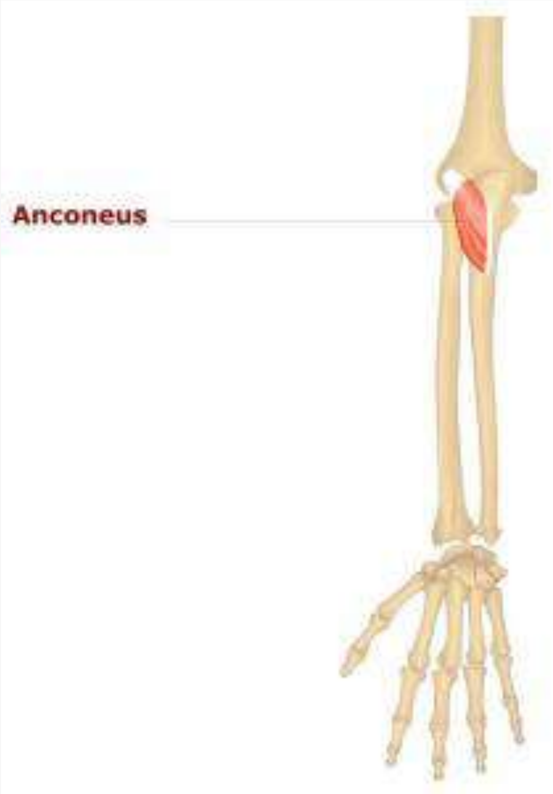


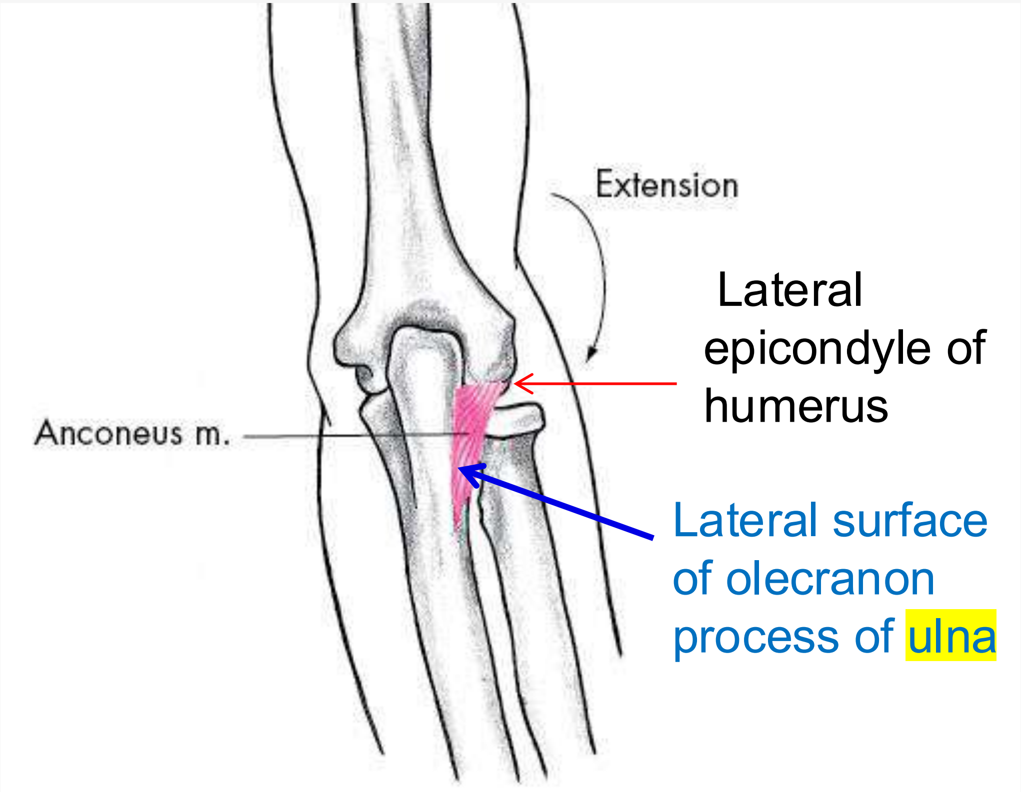
Extensor expansion of little finger

Action:

-Extension of metacarpophalangeal & Interphalangeal joints of little finger

Anconeus





Action:

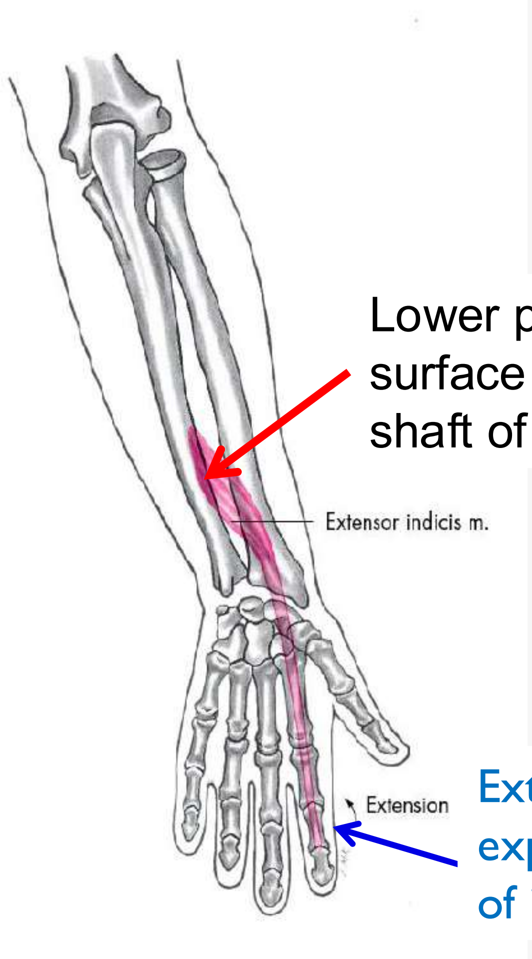
- Extension of elbow joint

Nerve supply:

Radial nerve

Posterior compartment Deep group

Extensor indicis



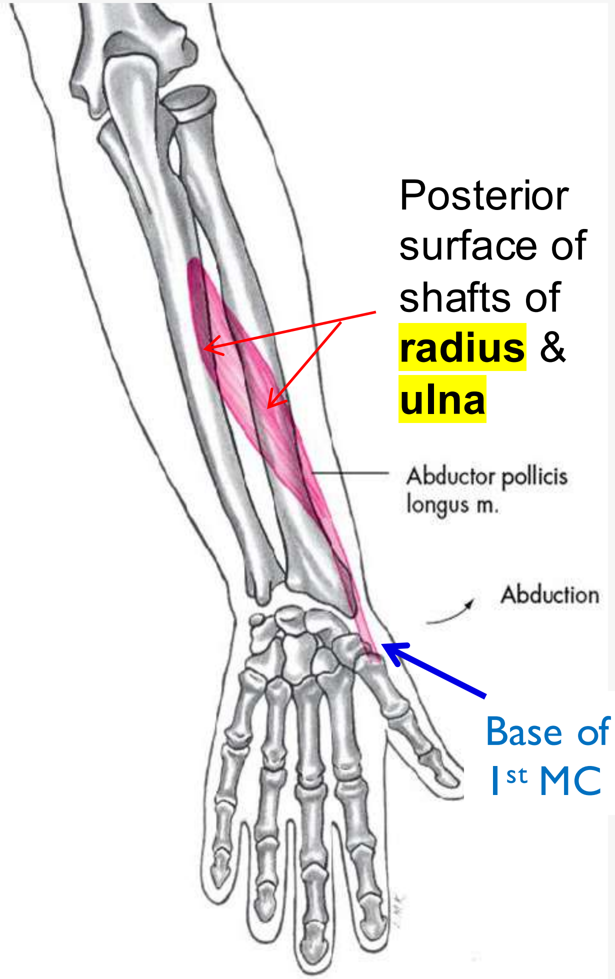
Lower posterior surface of shaft of **ulna**

Extensor expansion of index finger

Action:
- Extends metacarpophalangeal & Interphalangeal joints of index finger

The index finger has its own separate extensor, to enable it to extend independently from other fingers.

Abductor pollicis longus



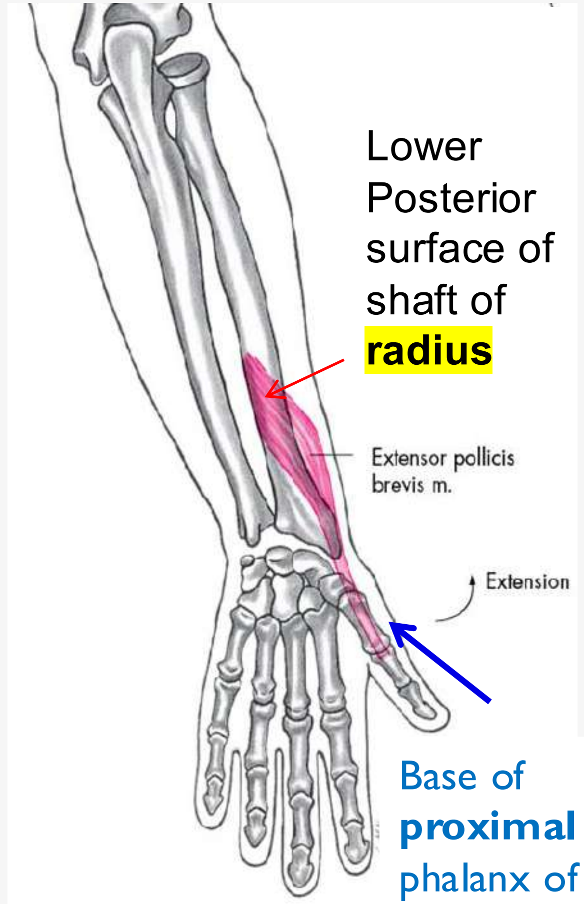
Posterior surface of shafts of **radius** & **ulna**

Base of Ist MC

Action:

- **Abducts** & extends 'thumb

Extensor pollicis brevis



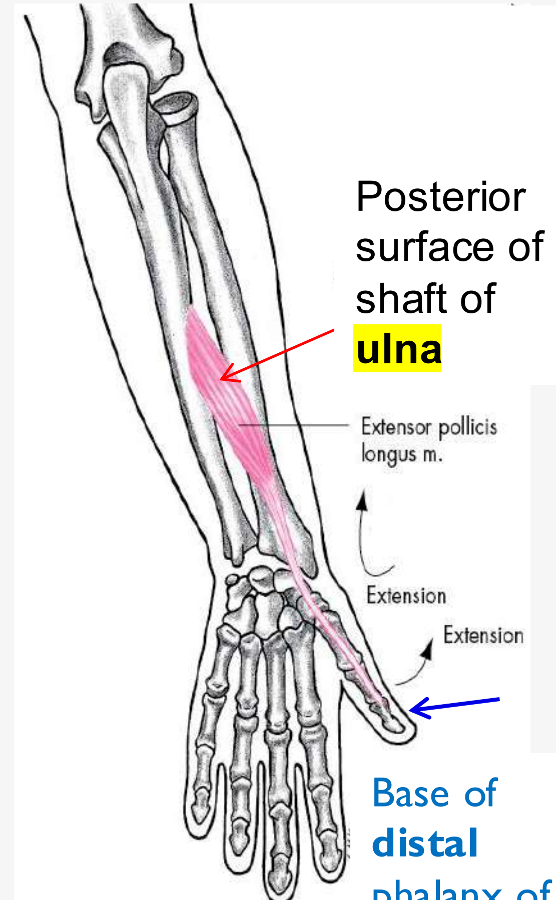
Lower Posterior surface of shaft of **radius**

Base of **proximal** phalanx of thumb

Action:

- Extends metacarpophalangeal joint of thumb / PP of thumb

Extensor pollicis longus



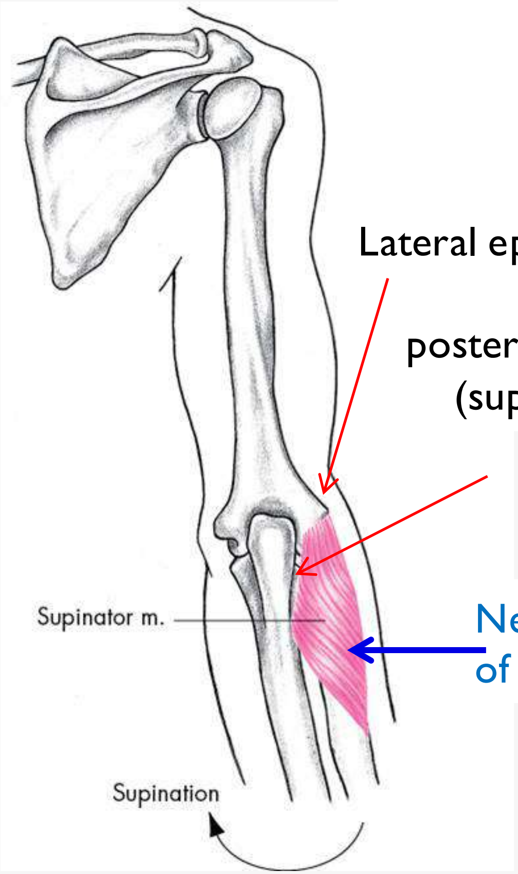
Posterior surface of shaft of **ulna**

Base of **distal** phalanx of thumb

Action:

-Extension of distal phalanx of thumb

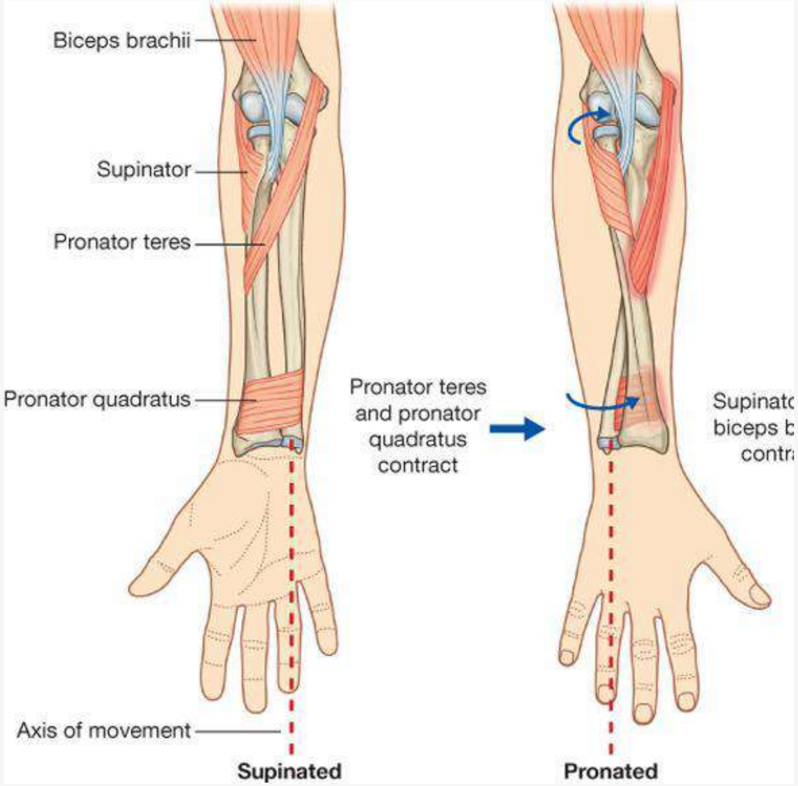
Supinator



Lateral epicondyle of **humerus**,
&
posterior upper part of **Ulna**
(supinator crest of ulna)

Neck & upper shaft of **radius**

Supinator & Pronator



Action:

Supination of forearm

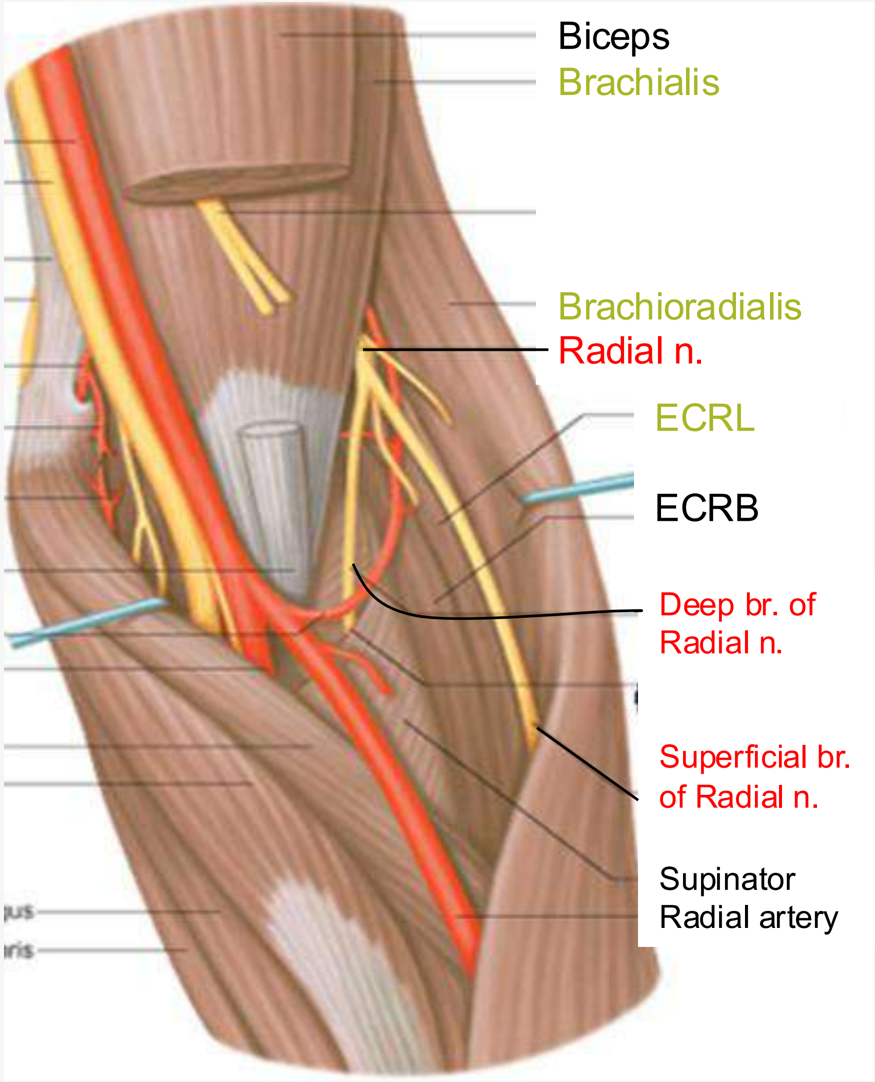
Radial Nerve ~ In the forearm

-It is the nerve of the **posterior & lateral** Compartments of the Forearm

~ It passes forward into the cubital fossa.

~ It passes downward **in front** of 'lateral epicondyle of 'humerus, between the brachialis (medially) brachioradialis & ECRL (laterally).

~ At the level of 'lateral epicondyle: it divides into superficial & deep branches.



Branches in the forearm:

■ **Muscular branches:** brachioradialis, ECRL, small branch to brachialis.+ anconeus.

■ Deep branch of the radial nerve:

~ It arises **in front** of ‘lateral epicondyle of ‘humerus
& enters ‘posterior comp. of ‘forearm, pierces the **supinator** muscle.

The deep branch of radial nerve continues as the posterior interosseous nerve and it descends with Posterior interosseous artery

~ Supplies: **Muscles** of the post. Compart. of ‘forearm
wrist & carpal **joints**.

■ Superficial branch of the radial nerve:

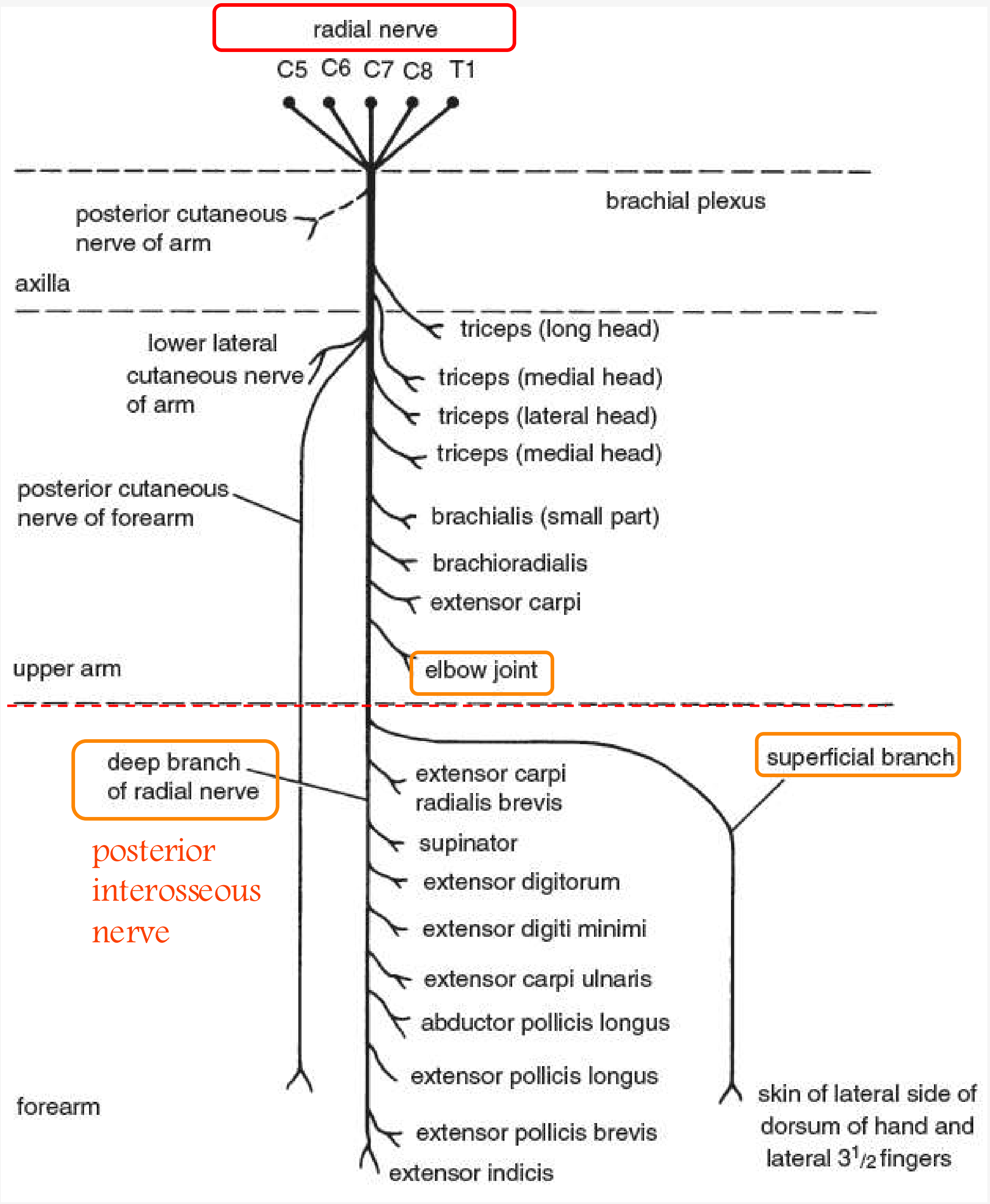
It is the direct continuation of radial nerve.

It is lateral to ‘radial artery.

~At distal part of ‘forearm:

it passes backward to reaches the posterior surface of the wrist, to supply the skin (see hand).

Radial Nerve



REMEMBER!!

The <u>Posterior</u> interosseous nerve
is the direct continuation of
Deep branch of Radial nerve

BUT
The <u>Anterior</u> interosseous nerve
is a branch of Median nerve

REMEMBER!!

The Anterior interosseous nerve
is a branch of Median nerve
With Anterior interosseous artery

BUT

The Posterior interosseous nerve
is the direct continuation of Deep branch of Radial nerve
With Posterior interosseous artery

~~~~~~~~~~~~

The Superficial branch of the radial nerve:
It is the direct continuation of Radial nerve.
With Radial artery
~~~~~~~~~~~~

ANATOMICAL SNUFF BOX

Remember … Muscles to the thumb ??

- Abductor pollicis longus AbPL
- Extensor pollicis brevis EPB
- Extensor pollicis longus EPL

It is a triangular skin depression on the lateral side of 'wrist.

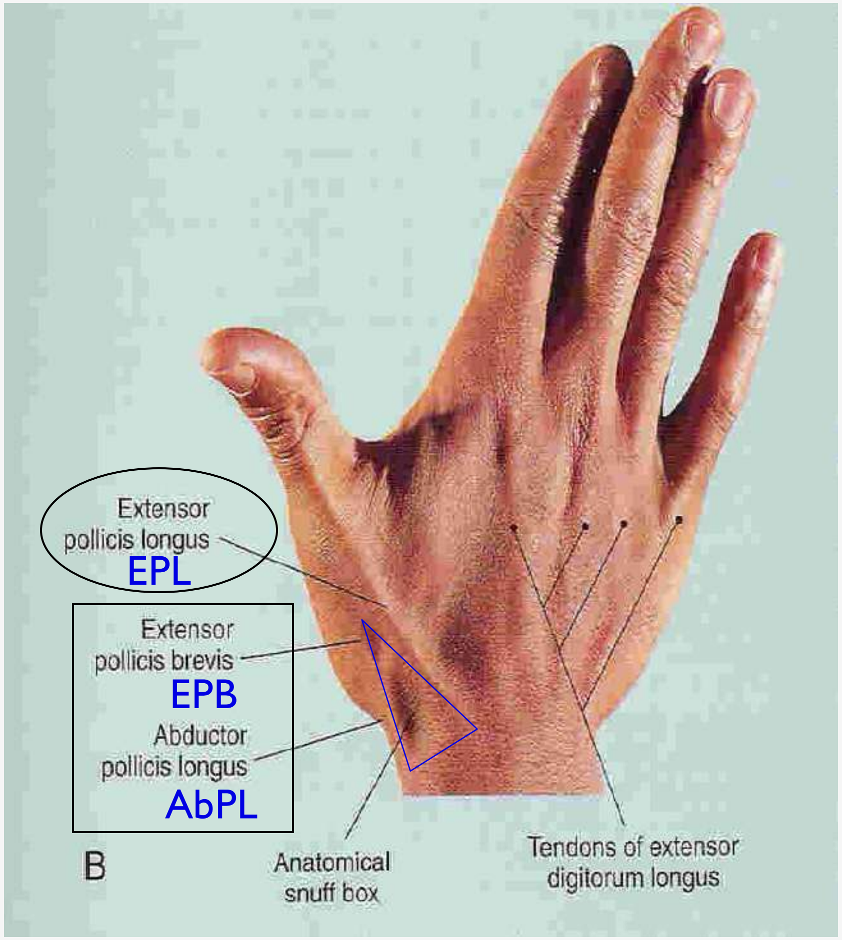
(appears when the thumb is abducted & extended)

Boundaries:

Laterally : AbPL & EPB tendons.
Medially : EPL tendon.
Floor : Scaphoid bone

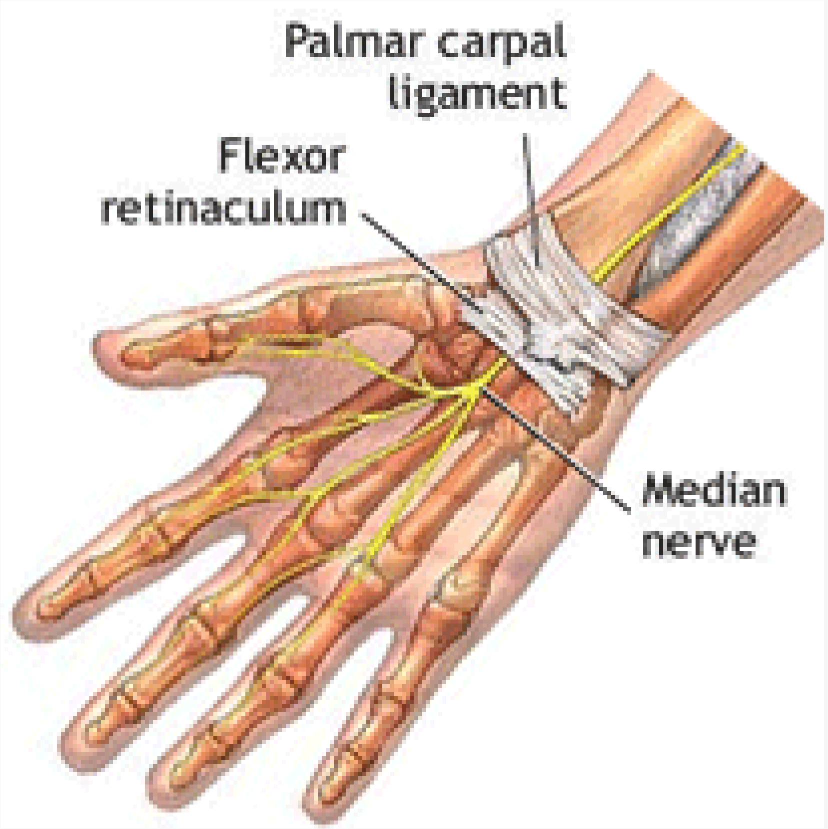
Clinical imp.

The site to feel the pulse of radial artery

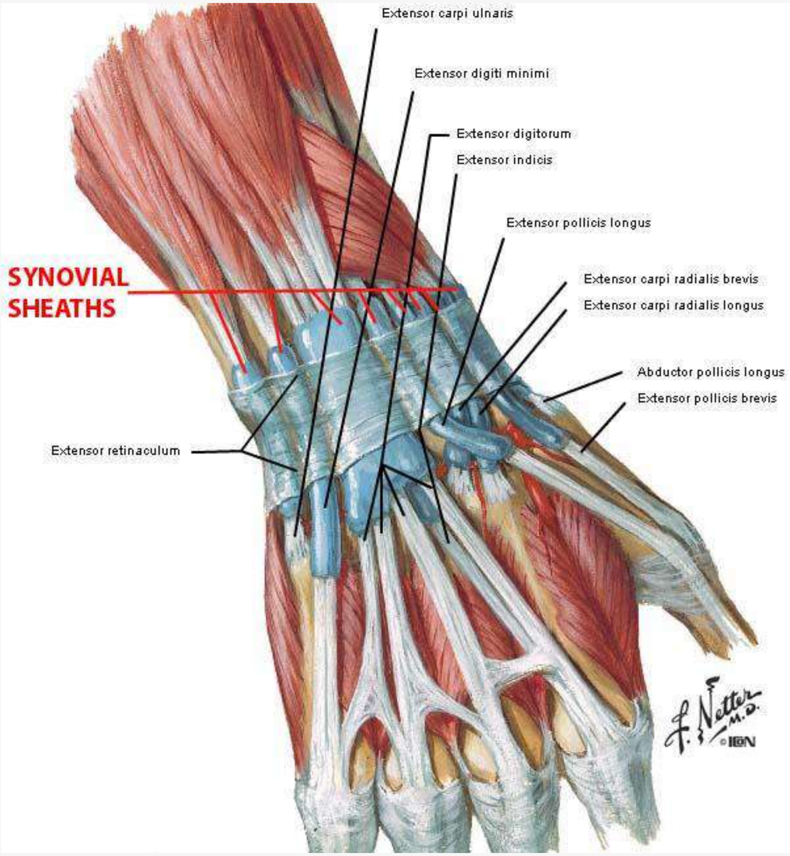


The retinacula

The flexor & extensor retinacula are strong bands of deep fascia that hold the long flexor & extensor tendons in position at the wrist.



Front (Flexor Ret.)



Back (Extensor Ret.)

The flexor retinaculum

A condensation of the deep fascia in front of the wrist joint & with the carpal bones it forms a tunnel, (carpal tunnel).

Attachments: …….

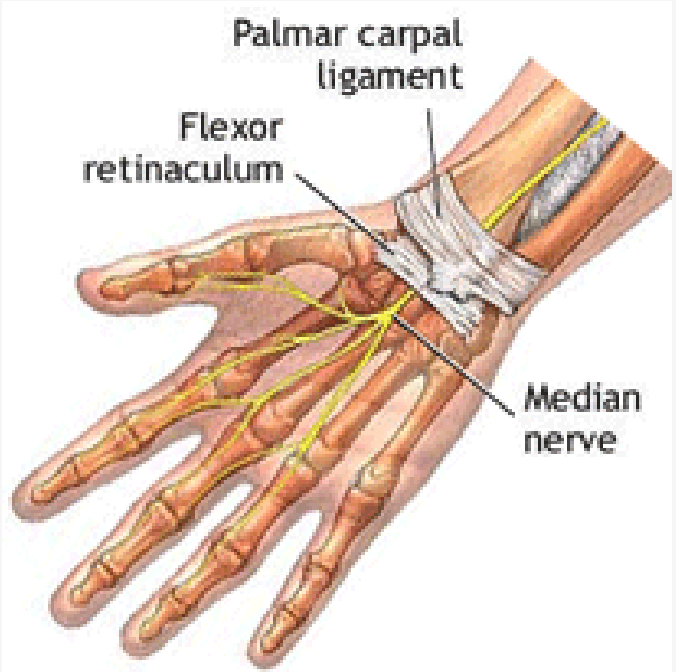
Structures passing deep & superficial to flexor retinaculum

Superficial:

Palmar cutaneous br. of Ulnar nerve
Palmar cutaneous br. of Median nerve
Palmaris longus tendon.
Ulnar artery &
Ulnar nerve

Deep:

Flexors digitorum superficialis tendons.
Flexor digitorum profundus tendons.
Median nerve.
Flexor pollicis longus tendon.
Flexor carpi radialis tendon.



The extensor retinaculum

A condensation of the deep fascia of the posterior aspect of the lower part of forearm and wrist joint.

-It stretches across the back of the wrist & converts the grooves on 'posterior surface of distal ends of Radius & Ulna into six separate tunnels, for the passage of the extensor tendons & to keep them in position.

Attachments:........

Structures passing deep & superficial to 'retinaculum

***Superficial*:**

Basilic & cephalic veins.

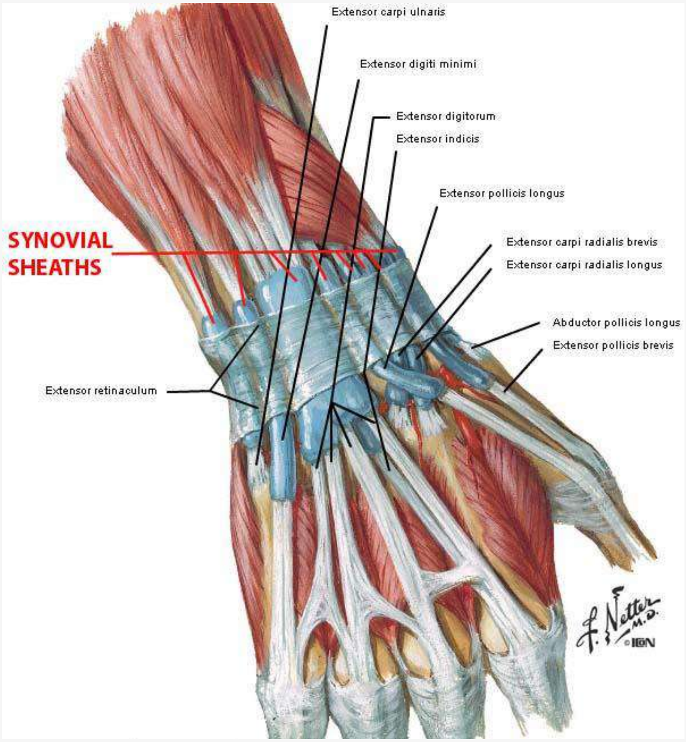
Superficial branches of Radial nerve

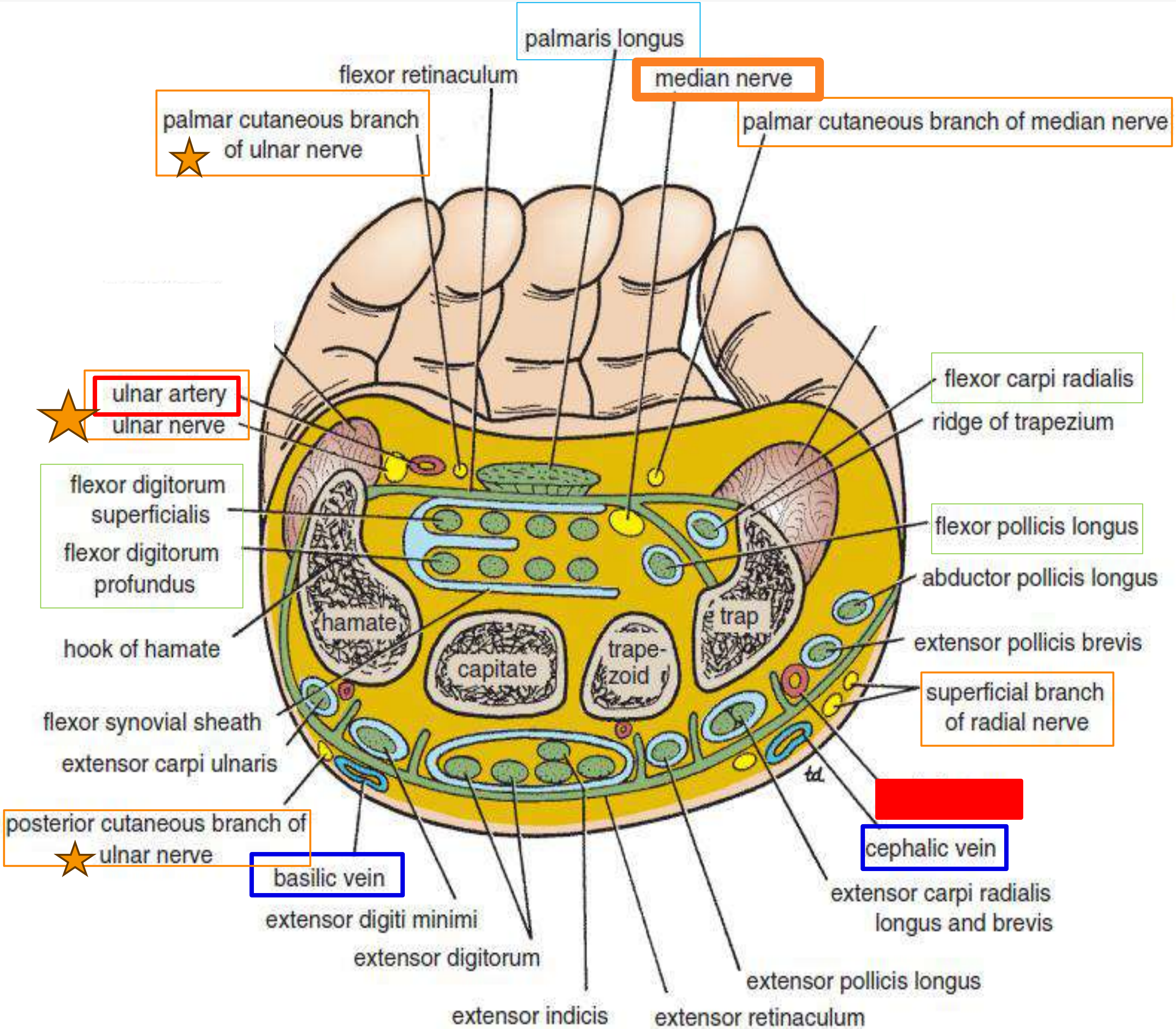
Posterior cutaneous branch of Ulnar nerve

Deep:

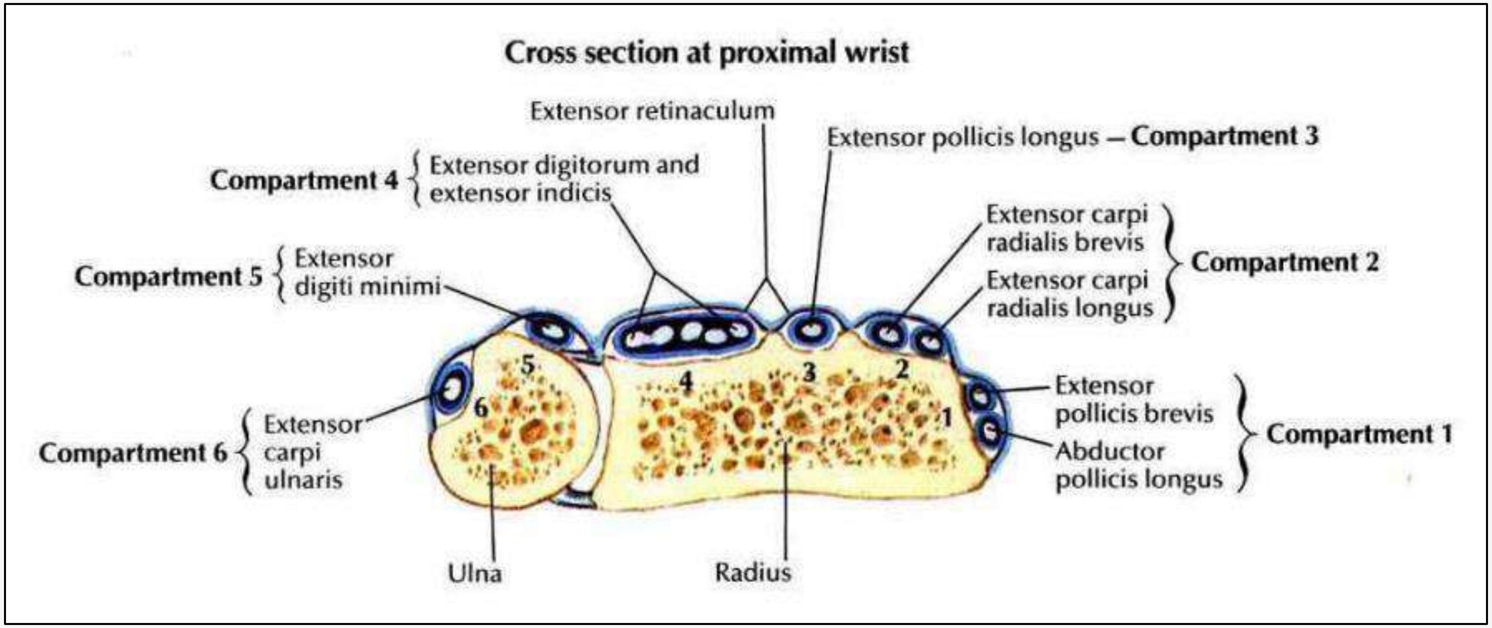
Extensor tendons….

Radial artery

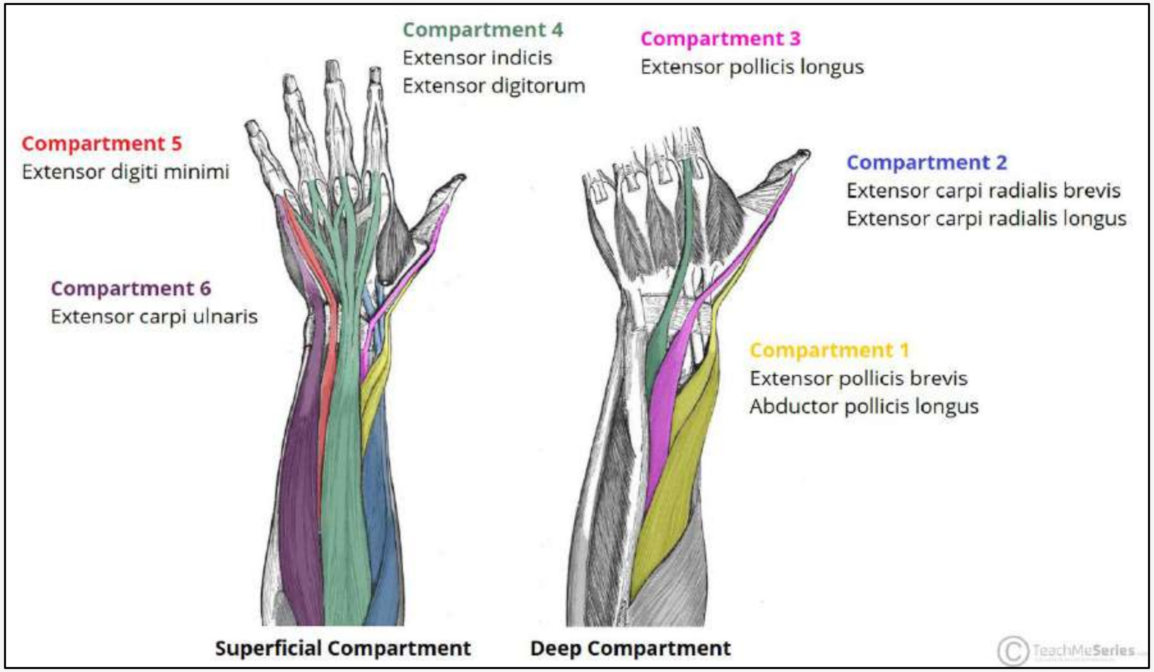



palmaris longus
flexor retinaculum
median nerve
palmar cutaneous branch of ulnar nerve
palmar cutaneous branch of median nerve
ulnar artery
ulnar nerve
flexor carpi radialis
ridge of trapezium
flexor digitorum superficialis
flexor digitorum profundus
flexor pollicis longus
abductor pollicis longus
hamate
trap
trape-zoid
capitate
hook of hamate
extensor pollicis brevis
superficial branch of radial nerve
flexor synovial sheath
extensor carpi ulnaris
td.
posterior cutaneous branch of ulnar nerve
basilic vein
cephalic vein
extensor carpi radialis longus and brevis
extensor digiti minimi
extensor digitorum
extensor pollicis longus
extensor indicis
extensor retinaculum

The extensor six compartments



Structures passing in six compartments




THANK YOU